

HEBBIAN SYNAPSES: Biophysical Mechanisms and Algorithms

*Thomas H. Brown, Edward W. Kairiss, and
Claude L. Keenan*

Department of Psychology, Yale University, New Haven,
Connecticut 06520

INTRODUCTION

The hypothesis that memory is associated with use-dependent synaptic modifications (cf. Tanzi 1893, Ramón y Cajal 1911, Freud 1894, Ziegler 1900, Woodworth 1921, Wood-Jones & Porteus 1928, Konorski 1948, Hebb 1949, Hayek 1952, Eccles 1983) has been carefully documented in higher invertebrates (Carew & Sahley 1986, Byrne 1987, Kandel et al 1987). Here we review the type of synaptic learning mechanism originally suggested by Hebb (1949) to operate in higher vertebrates. Theoretical studies suggest that useful and potentially powerful forms of learning and self-organization can emerge in networks of elements that are interconnected by various formal representations of a Hebbian modification (Palm 1982a, Kohonen 1984, 1987, Anderson 1985, Denker 1986, Hopfield & Tank 1986, Linsker 1986, 1988a,b, Tesauero 1986, Bear et al 1987, Pearson et al 1987, Klopff 1988, 1989, Schmajuk & Moore 1988, Sejnowski & Tesauero 1989).

Interest in the computational power of networks of elements that are interconnected by Hebbian modification algorithms has been enhanced by the recent neurophysiological discovery of a synaptic mechanism in the hippocampus that produces modifications that resemble some contemporary interpretations of Hebb's original postulate for learning (Kelso et al 1986, Malinow & Miller 1986, Sastry et al 1986, Wigström et al 1986). This mechanism is responsible for a type of long-term synaptic potentiation

(LTP) that can be experimentally induced in parts of the hippocampus and probably in many other brain regions (Brown et al 1988a,b).

In this review we first summarize the evolution of the current concept of a Hebbian synaptic learning mechanism and propose a contemporary definition. This naturally leads to the question of whether Hebbian synapses have now been demonstrated to exist. We approach this question by reviewing recent facts and hypotheses about LTP that are pertinent to our definition of a Hebbian synaptic modification. At least one type of LTP that has been demonstrated *in vitro* in the hippocampus appears to qualify as a specific instance of a Hebbian modification. After reviewing some additional examples and variations of a Hebbian synaptic modification, we turn to theoretical analyses of Hebbian learning rules. Several of the modification algorithms that have been considered in studies of adaptive neural networks also satisfy our definition of a Hebbian modification. Additional experimental and theoretical research will be required to understand the conditions under which these or other algorithms can be viewed as useful abstractions of the neurophysiology of LTP.

THE IDEA OF A HEBBIAN SYNAPSE

The idea that associative learning emerges from a Hebb-like conjunctive or correlational mechanism at the synapse has a history that can be traced back a century. Here we present highlights of the evolution of the current concept.

The Law of Neural Habit

An antecedent to the idea of a Hebbian synapse is found in James (1890; reprinted in Anderson & Rosenfeld 1988). James argued that the laws of association reflect the laws of cerebral physiology (p. 225):

How does a man come, after having the thought of A, to have the thought of B the next moment? or how does he come to think of A and B always together? These were the phenomena which Hartley undertook to explain by cerebral physiology. I believe he was, in essential respects, on the right track, and I propose simply to revise his conclusions by the aid of distinctions which he did not make.

In particular, James maintained that: "there is no other elementary causal law of association than the law of neural habit: When two elementary brain processes have been active together or in immediate succession, one of them, on reoccurring, tends to propagate its excitement into the other" (p. 226). James' law of neural habit thus indicates the basic conditions for change in the form of a qualitative statement about the activity-modification relationship. Anderson & Rosenfeld (1988) note that when

the term “neurons” is substituted for “elementary brain processes,” the habit law immediately suggests a set of learning rules that include the idea of a Hebbian synapse.

The Synaptic Hypothesis for Learning

James did not discuss the subcellular locus and nature of the neuronal mechanism that causes coactivated “brain processes” to change. These issues were addressed by Tanzi (1893), who identified the synapse as the locus of the modification. The proposed nature of the modification involved changes in the strengths of previously existing synaptic connections, an idea that remains the best-established hypothesis today (Abrams & Kandel 1988). A relatively clear statement of the synaptic hypothesis for learning appears in Wood-Jones & Porteus (1928). In discussing synaptic “linkages” onto “dendrons,” these authors concluded (p. 48) that “the power which creates these linkages must be plastic,” and they suggested a biophysical mechanism for the expression of this plastic change (pp. 355–56):

A first step toward an explanation is the recognition of the importance of the synapse, the break in the nervous chain, at the point of contact between the fibres of the affecter and the effector system. We must assume that there is a certain resistance at this break to the passage of the nervous impulse, a resistance that is capable under certain conditions of being lowered so as to either permit the passage of the impulse unimpeded or to step it down, so to speak, so that it will not call forth any, or only a very weak response in the muscles. The lowering of the synaptic resistance we have called facilitation.

This explanation of learning in terms of synaptic facilitation has a more modern sound if rephrased in terms of an increase in the synaptic conductance instead of a lowering of the synaptic resistance.

The Conditions for Synaptic Modification

Wood-Jones & Porteus (1928) did not elaborate on the conditions that trigger or induce the enhanced synaptic efficacy. Hebb (1949) took up this matter two decades later (see also Konorski 1948) in his famous neuropsychological treatise, which contains a statement now known (Stent 1973) as “Hebb’s postulate of learning” (p. 62):

When an axon of cell A is near enough to excite cell B or repeatedly or consistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A’s efficiency, as one of the cells firing B, is increased.

Hebb proposed this change as the basis of a memory formation and storage process that would cause enduring modifications in the elicited activity patterns of spatially distributed “nerve cell assemblies.” This postulate combines the law of neural habit with the synaptic hypothesis for learning. It specifies the location of the modification and it provides a qualitative

statement of the conditions for change. It does not furnish much guidance in regard to the quantitative details of the activity-modification relationship (the modification rule or algorithm). In regard to the nature of the biophysical mechanism that increases synaptic efficiency, Hebb (1949) suggested a decrease in the synaptic resistance.

The Concept of a Hebbian Synaptic Modification

Since 1949 the concept of a Hebbian synapse has evolved to include several key features, which are emphasized to a greater or lesser extent by different investigators (Stent 1973, Sejnowski 1981, Palm 1982a, Kohonen 1984, 1987, Anderson 1985, Levy & Desmond 1985, Denker 1986, Kelso et al 1986, Tesauro 1986, Bear et al 1987, Klopff 1988, 1989, Levy & Burger 1987, Brown et al 1988a,b, 1989, Reiter & Stryker 1988, Sejnowski & Tesauro 1989, Schmajuk & Moore 1988). These key features form the basis of the contemporary understanding of a Hebbian synaptic mechanism. We define a Hebbian synapse as one that uses a *time-dependent, highly local, and strongly interactive mechanism to increase synaptic efficacy* as a function of the conjunction or correlation between pre- and postsynaptic activity. The key terms in this definition are elaborated below.

TIME-DEPENDENT MECHANISM Modifications in a Hebbian synapse depend on the exact time of occurrence of pre- and postsynaptic activity. In particular the modifications are driven by what Klopff (1988) calls a “real-time mechanism” (cf. Sutton & Barto 1981, Donegan & Wagner 1987), which he defines as follows (pp. 17–18):

Real-time learning mechanisms emphasize the temporal association of signals: each critical event in the sequence leading to learning has a time of occurrence associated with it and this time plays a fundamental role in the computations that yield changes in the efficacy of synapses. It should be noted that “real-time,” in this context, does not mean continuous time as contrasted with discrete time nor does it refer to a learning system’s ability to accomplish its computations at a sufficient speed to keep pace with the environment within which it is embedded. Rather, a real-time learning mechanism, as defined here, is one for which the time of occurrence of each critical event in the sequence leading to learning is of fundamental importance with respect to the computations the learning mechanism is performing. Real-time learning mechanisms may be contrasted with nonreal-time learning mechanisms such as the perceptron . . . , adaline . . . , or back propagation . . . learning mechanisms for which error signals follow system responses and only the order of the inputs, outputs, and error signals is important, not the exact time of occurrence of each signal, relative to the others.

Because the expression *real-time mechanism* may have unintended connotations, we call this a *time-dependent mechanism*, but the meaning is the same.

LOCAL MECHANISM The synapse is the transmission site where the signals or information representing ongoing activity in the pre- and postsynaptic elements are in spatiotemporal contiguity. A Hebbian synapse uses this locally available information to cause a local, input-specific synaptic modification. The necessary information or signal for change can be nothing more than the natural consequence of ordinary intercellular communication at the synapse. Hebbian synapses are thus said to enable an “unsupervised” form of learning—in the sense that a specific external “teacher” signal is not required to instruct change on an individual, synapse-by-synapse, basis. The idea of a local mechanism does not exclude some form of neuromodulatory control over the modification process. A “reinforcement signal” is not excluded.

INTERACTIVE MECHANISM Whether a change occurs at a Hebbian synapse depends on activity levels on both sides of the synaptic cleft. Palm (1982a) notes that a Hebbian mechanism is based on a “true interaction” between pre- and postsynaptic activity. “Noninteractive” mechanisms or rules can be purely presynaptic, purely postsynaptic, or a superposition of these, but there is no true interaction. Palm’s (1982a) distinction between interactive and noninteractive rules is similar to Finkel & Edelman’s (1987) distinction between “dependent” and “independent” rules (p. 712):

In an independent synaptic rule, presynaptic and postsynaptic modifications can occur independently, a reasonable property given the individuality of the two neurons involved. Nevertheless, since the work of Hebb (1949) and von [sic] Hayek (1952), most theoretical models have opted for some form of dependent synaptic rule. In a dependent rule, the locus of the change in synaptic efficacy can be presynaptic and/or postsynaptic (and usually is not specified), but the change is contingent upon events at both loci.

We would add only that this dependence or interaction can be statistical and that the magnitude of the statistical interaction can vary across types of synapses or physiological conditions. A noninteractive mechanism is thus one in which either there is no interaction or the statistical interaction is inconsequential or undetectable.

CONJUNCTIVE OR CORRELATIONAL MECHANISM We have not yet addressed the logical or statistical form of the interactive mechanism (cf. Palm 1982a). One interpretation of Hebb’s postulate is that the condition for change is simply the conjunction of pre- and postsynaptic activity. Thus the co-occurrence of (some level of) pre- and postsynaptic activity (within some short time interval) is sufficient to cause the synaptic enhancement. For this reason, a Hebbian synapse is sometimes called a *conjunctional synapse*. Did Hebb (1949) mean for the coincidence of pre- and postsynaptic activity to be sufficient to cause a modification? Or was there also an implied

requirement for a positive correlation between pre- and postsynaptic activity? Hebb was not clear on this point. In Sejnowski's (1977a,b) formalization of Hebb's postulate, correlation over time between pre- and postsynaptic activity is responsible for changes in synaptic efficacy. This condition would be satisfied if cell A "consistently takes part in firing" (Hebb 1949) cell B. Thus a Hebbian synapse also is sometimes called a *correlational synapse* (Anderson 1985). Sejnowski (1977a,b) expresses the condition for change mathematically in terms of the covariance between pre- and postsynaptic activity. A similar idea can also be expressed in terms of conditional probabilities.

Generalizing the Concept of a Hebbian Modification

SYNAPTIC ENHANCEMENT AND DEPRESSION Hebb (1949) did not discuss the consequence of uncorrelated or negatively correlated pre- and postsynaptic activity. Are we to assume that uncorrelated activity reduces synaptic efficacy? Or that it has no effect on efficacy? Note that the definition of a Hebbian synapse given above does not exclude additional processes that can decrease synaptic strength. The idea that positively correlated activity causes synaptic strengthening and that either uncorrelated or negatively correlated activity causes synaptic weakening was in fact an essential part of Stent's (1973) extension of the Hebbian idea to the problem of understanding experience-dependent aspects of the development of striate cortex (cf. Bear et al 1987, Reiter & Stryker 1988, Singer 1988).

Later we consider two categories of modification rules: those that can only strengthen the synapse and those that can strengthen or weaken the synapse, depending on the relationship between pre- and postsynaptic activity levels. The latter may be considered extensions or *generalizations of the original concept* of a Hebbian modification as defined above. Palm (1982a,b) has suggested a formalism for classifying what he calls "Hebbian," "anti-Hebbian," and "non-Hebbian" synaptic modifications. According to this scheme, a Hebbian synapse increases its strength with correlated pre- and postsynaptic activity and decreases its strength with negatively correlated activity. Conversely, an anti-Hebbian synapse rewards negatively correlated activity and punishes correlated activity. In both cases, modification of synaptic efficacy involves a real-time, local, and interactive mechanism. A non-Hebbian synapse is one that does not rely on a real-time, local, and interactive mechanism. At such "non-interactive" synapses, the modifications can be expressed as simple superpositions of purely presynaptic or purely postsynaptic rules. Palm (1982a,b) has discussed vector representations of these synaptic modification rules.

We should point out in passing that the term "anti-Hebbian" is not used

consistently in the literature. As just noted, for Palm (1982a), an anti-Hebb rule is one that decreases synaptic efficacy as a result of positively *correlated* pre- and postsynaptic activity. Callaway et al (1987) use the term anti-Hebbian in this sense to describe changes at neuromuscular junction during development (see also Callaway et al 1989). Kohonen & Oja (1976) use this form of rule to construct an orthogonalizing filter, and Hopfield et al (1983) use an anti-Hebb rule to remove spurious memory states. In contrast, Levy & Desmond (1985) use the phrase Hebb/anti-Hebb to refer to a rule (discussed below) that increases synaptic efficacy when pre- and postsynaptic activity are coincident, and decreases synaptic efficacy if postsynaptic activity is unaccompanied by presynaptic activity (see also Ranck 1964, Rosenblatt 1967, Stent 1973, Kohonen et al 1974, Cooper et al 1979).

As noted above, the idea of a Hebbian modification can be generalized to include the combination of an interactive synaptic enhancement and some type of activity-dependent synaptic depression. We refer to this combination as a *generalized Hebbian synaptic mechanism*. The depression can be of the noninteractive type or any of several different types of interactions between pre- and postsynaptic activity (cf. Stent 1973, Sejnowski 1977a,b, Palm 1982a, Kohonen 1984, Linsker 1986, Bear et al 1987, Levy & Burger 1987). The interactive condition that causes depression can simply be non-coincident pre- or postsynaptic activity. By this we mean activity that occurs on one side of the synaptic cleft but not the other. In this type of synapse, the occurrence of (some level of) presynaptic activity in the absence of (some level of) postsynaptic activity (Figure 1D) or the exact opposite activity relationship (Figure 1C) causes synaptic depression. Alternatively, the condition that causes synaptic weakening can be the occurrence of pre- and postsynaptic activity that is statistically uncorrelated or that is negatively correlated in time.

TEACHER SIGNALS AND REINFORCEMENT LEARNING Although a Hebbian mechanism uses only local information, the modification process may be subject to global control signals. Such signals may enable the induction or consolidation of changes at synapses that have met the above criteria for a Hebbian modification. A global "reinforcement signal" can thus control Hebbian plasticity in a large population of activated synapses. Substances that have been demonstrated to act as neuromodulators (such as catecholamines and acetylcholine) are plausible candidates for this role. Such global modulation of a local process is different from an external "teacher" signal that explicitly "instructs" selective modification on a synapse-by-synapse basis independent of local activity.

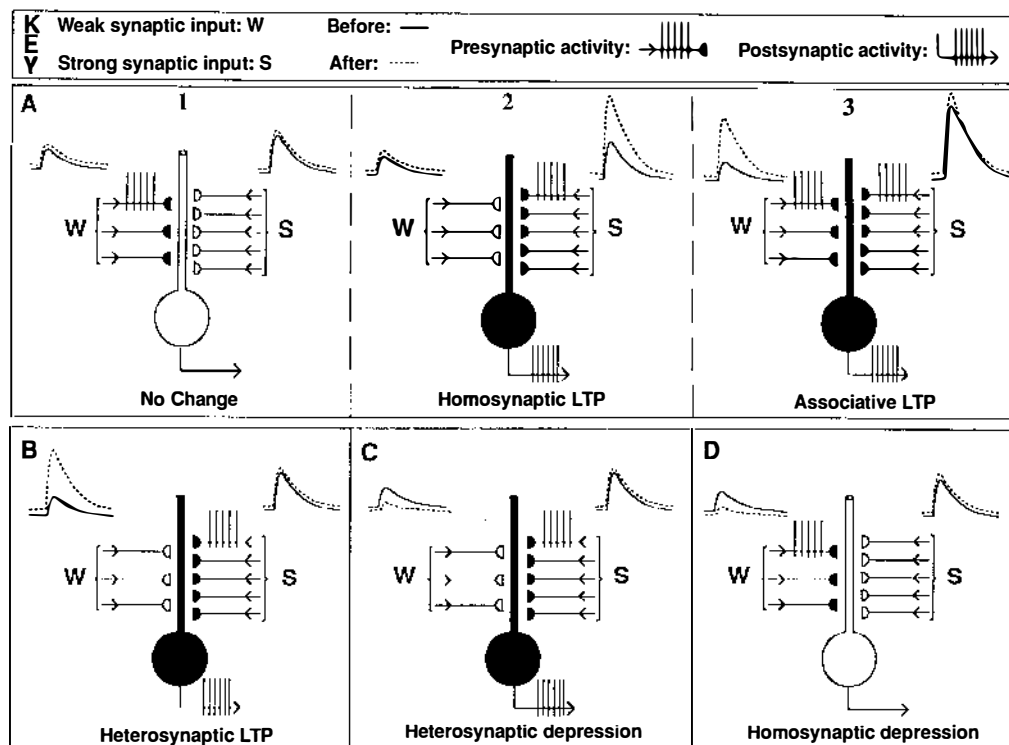


Figure 1 Illustration of several varieties of use-dependent synaptic changes. Each neuron is shown receiving two sets of nonoverlapping synaptic inputs, one weak (*W*) and the other strong (*S*). The difference between these two sets reflects the number of afferent fibers (for a quantitative interpretation, see Brown et al 1988). The waveforms above each input illustrate schematically the excitatory postsynaptic potential produced by a single stimulation of that input before (*solid curve*) and after (*broken curve*) tetanic (high-frequency) stimulation of one or both inputs. *Filled elements* indicate activity during the conditioning (tetanic) stimulation. **A. Associative LTP.** *A1.* Tetanic stimulation of *W* alone does not cause LTP in either input. *A2.* Subsequent tetanic stimulation of *S* may cause homosynaptic LTP (synaptic enhancement in the stimulated input) but not heterosynaptic LTP (enhancement in the unstimulated input). *A3.* Concurrent tetanic stimulation of *W* and *S* causes associative LTP (enhancement in *W*). Further, LTP in the *S* input may also occur under these conditions (but is not illustrated). **B. Heterosynaptic LTP.** Tetanic stimulation of the *S* input alone causes LTP in the unstimulated *W* input. Homosynaptic LTP in the stimulated *S* input may also occur under these circumstances (but is not shown). **C. Heterosynaptic depression.** Stimulation of the *S* input alone causes depression in the unstimulated *W* input. Homosynaptic LTP in the *S* input may also occur under these conditions (but is not shown). **D. Homosynaptic depression.** High frequency stimulation of *W* alone causes depression in the *W* input. In principle, homosynaptic depression could also occur in the *S* input if tetanic stimulation of this input did not cause homosynaptic LTP.

HEBBIAN SYNAPTIC MECHANISMS

Hebbian synapses have now been shown to exist in the sense that one can demonstrate experimentally a use-dependent form of synaptic enhancement that is governed by a time-dependent, highly local, and strongly interactive mechanism. This mechanism appears to be responsible for one form of hippocampal LTP. Whether this potential for a Hebbian synaptic modification is actually used in the development and organization of behavior through the formation of Hebb's (1949) nerve "cell assemblies" remains to be seen.

Varieties of Hippocampal Long-Term Potentiation

LTP is a use-dependent and persistent increase in synaptic strength that can be induced by brief periods of synaptic stimulation (reviewed in Teyler & DiScenna 1987, Bliss & Lynch 1988, Brown et al 1988a,b, 1989). LTP was first described in the rabbit hippocampal formation (Bliss & Lømo 1973, Bliss & Gardner-Medwin 1973) and was subsequently demonstrated to occur in numerous excitatory synapses of the central and peripheral nervous systems of vertebrates and invertebrates (Brown & McAfee 1982, Racine et al 1983, Baxter et al 1985, Briggs et al 1985, Miller et al 1987, Walters & Byrne 1985, Teyler & DiScenna 1987, Artola & Singer 1987, Bindman et al 1988).

There are at least two and possibly many more different LTP mechanisms or forms of LTP (Brown et al 1989). This conclusion is based on the kinetics of the passive decay of the enhancement, the activity-modification relationships that govern the induction and expression of the enhancement, and the particular membrane receptors that control the modification process (cf. Barrionuevo & Brown 1983, Baxter et al 1985, Briggs et al 1985, Racine & Kairiss 1987, Brown et al 1988b, 1989, Racine & deJonge 1988, Johnston et al 1989, Cotman et al 1988). Most of our knowledge of the cellular neurophysiology of LTP in the hippocampus comes from two synaptic systems: the Schaeffer collateral/commissural (Sch/comm) inputs to the CA1 pyramidal neurons and the mossy-fiber synaptic inputs to the CA3 pyramidal neurons [reviewed in Brown et al (1989)].

The Sch/comm synapses display an associative form of LTP (Barrionuevo & Brown 1983, Kelso & Brown 1986) (see Figure 1A, panels 1–3) that is governed by an interactive mechanism (Figure 2). The enhancement depends on both presynaptic activity and the voltage across the postsynaptic membrane (Kelso et al 1986, Malinow & Miller 1986, Wigström et al 1986, Gustafsson et al 1987, Brown et al 1989). The actual statistical strength of the interaction has not been measured at these or any other

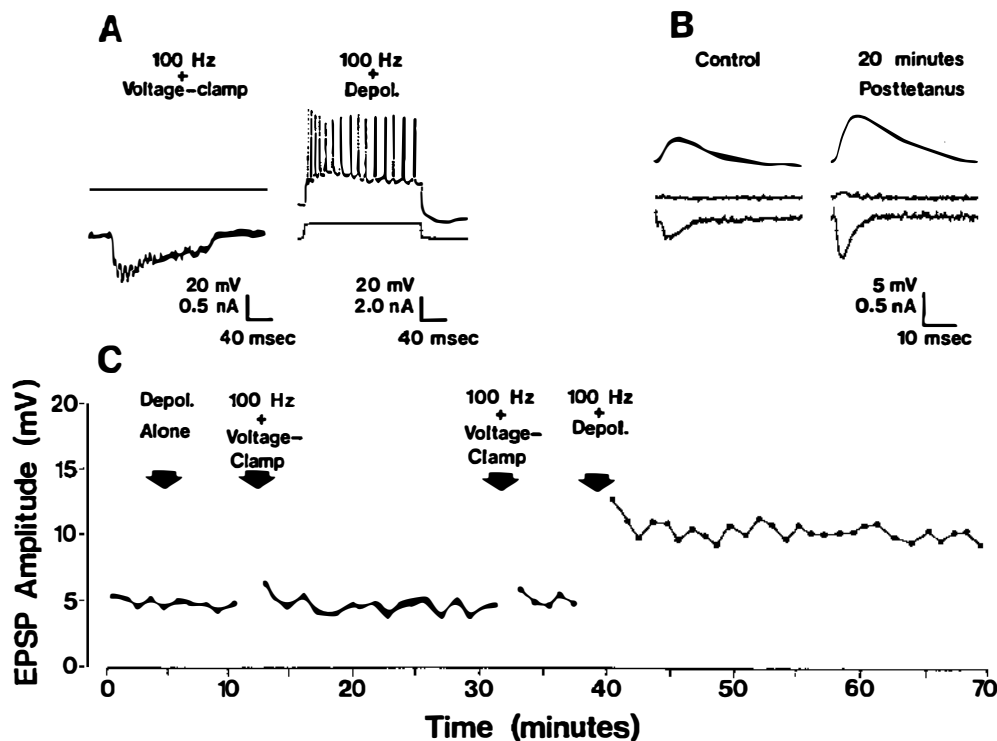


Figure 2 Direct demonstration of the interactive mechanism. Recordings are of Schaffer collateral synaptic responses in hippocampal neurons of region CA1 (see Figure 3A). *A.* (Left) Voltage-clamp record of inward synaptic currents (lower trace) and membrane potential (upper trace) during the synaptic stimulation train. (Right) Current-clamp recording of postsynaptic action potentials (upper trace) produced by an outward current step (lower trace) that is paired with the synaptic stimulation train. *B.* Current-clamp (top traces) and voltage-clamp (bottom traces) records before and 20 min after pairing synaptic stimulation with the outward current step. Middle trace is the membrane potential during voltage clamp. *C.* EPSP amplitudes as a function of the time of occurrence (arrows) of three manipulations: an outward current step (depol. alone) or synaptic stimulation trains delivered while applying either a voltage clamp (100 Hz + voltage clamp) or an outward current step (100 Hz + depol.). Each point is the average of five consecutive EPSP amplitudes. Modified from Kelso et al (1986).

plastic synapses. No published data indicate whether LTP in the mossy-fiber synapses involves a strictly noninteractive mechanism, a weakly interactive mechanism, or a strongly interactive mechanism. The spatio-temporal features of the activity-enhancement relationships have not been studied in the mossy-fiber system. Most of the rest of the discussion is therefore restricted to the associative type of LTP displayed by the Sch/comm synapses (see Figure 3A for the experimental preparation).

Some Features of Associative LTP in Hippocampus

INTENSITY THRESHOLD Bliss & Gardner-Medwin (1973) noted that the intensity of the electrical shocks delivered in a conditioning train (also called a *tetanic stimulation*) is an important determinant of LTP induction in the perforant pathway input to the dentate gyrus. Low-intensity tetanic stimulations were generally ineffective for inducing LTP. Increasing the stimulus intensity increased the effectiveness of the tetanic stimulation. This stimulus intensity effect has been called *cooperativity* (McNaughton et al 1978). As we use the term below, *cooperativity* refers to a condition in which the probability of inducing LTP in a particular set of synapses by some presynaptic stimulation pattern, or the magnitude of the enhancement resulting in this set of synapses from this stimulation pattern, is known or suspected to be a monotonic increasing function of the total number of synaptic inputs that are stimulated at about the same time.

This shock-intensity effect has been seen at two of the three most commonly studied synaptic systems of the hippocampal formation: the perforant pathway inputs to the dentate gyrus (Bliss & Gardner-Medwin 1973, McNaughton et al 1978) and the Sch/comm inputs to the CA1 region (Barrionuevo & Brown 1983, Lee 1983). In these systems, stimulation of a small number of afferent inputs [which produce a weak (*W*) postsynaptic response] commonly fails to induce LTP in that set of synapses (Figure 1A, panel 1), whereas the identical stimulation pattern applied to a larger total number of afferents [which collectively produce a strong (*S*) postsynaptic response] does induce LTP in that larger set of inputs (Figure 1A, panel 2). This intensity effect has not yet been reported to be a feature of the mossy-fiber synaptic inputs to the CA3 region.

SPATIOTEMPORAL SPECIFICITY *Associative LTP* (Levy & Steward 1979, Barrionuevo & Brown 1983, Kelso & Brown 1986) refers to a particular type of interaction between separately and independently stimulated *W* and *S* synaptic inputs to a neuron or small region (see Figure 3A for experimental setup). The nature of the interaction that defines associative LTP is taken from the perspective of the *W* input. In particular, associative LTP is an enhancement that can be induced in the *W* input if it and the *S*

input are stimulated together at about the same time, but not if it and the *S* input are stimulated separately at very different times or if only one of them is stimulated. Figure 1*A* illustrates associative LTP graphically (panels 1–3). Note that the definition of associative LTP specifically excludes the possibility of heterosynaptic LTP (Figure 1*B*) produced in the *W* input by tetanic stimulation of the *S* input. A synaptic system that displays associative LTP will exhibit cooperativity, but the reverse need not be true.

The activity-enhancement relationships that govern associative LTP (Figure 1*A*) in the hippocampus are relevant to its possible role as a synaptic substrate for aspects of learning. Some key spatiotemporal features of the activity-enhancement relationships include the following (Kelso & Brown 1986): (*a*) the induction of the functional modulation is rapid, (*b*) the expression of the enhanced synaptic strength is persistent, (*c*) the modification of one synaptic input can be conditionally controlled by temporal contiguity with activity in another synaptic input to the same region, and (*d*) the associative enhancement appears to be specific to just those synapses whose activity conforms to the temporal requirement.

These features are shared by the synaptic mechanisms that have been shown in *Aplysia* to underlie simple forms of associative memory (Carew & Sahley 1986, Byrne 1987, Kandel et al 1987), although there are well-known differences in the synaptic mechanisms in the two systems (cf. Carew et al 1984, Kelso et al 1986). Sejnowski & Tesauro (1989) describe different implementations of a Hebbian algorithm, one of which is functionally equivalent to parts of the gill-withdrawal circuitry in *Aplysia*, while another resembles the LTP mechanism described below.

Biophysical Models of LTP Induction in Hippocampus

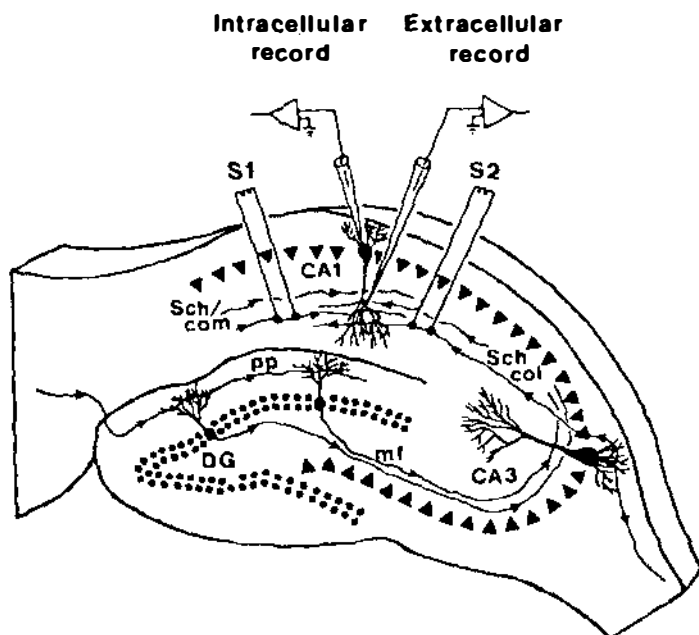
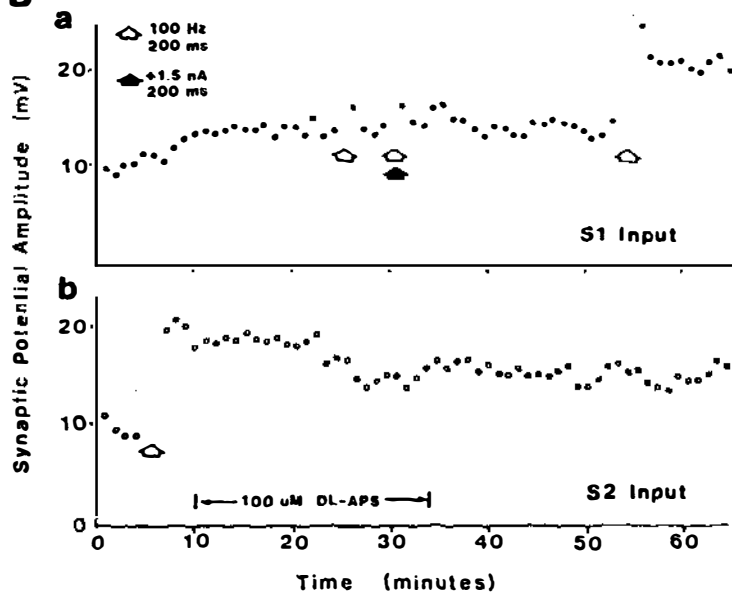
The spatiotemporal features of associative LTP (cf. Barrionuevo & Brown 1983, Kelso & Brown 1986, Levy & Desmond 1985, Levy & Burger 1987, Brown et al 1989) can be accounted for easily by a Hebb-like mechanism. Four research groups furnished evidence for an interactive mechanism underlying the induction of LTP in the Sch/comm synapses (Kelso et al 1986, Malinow & Miller 1986, Sastry et al 1986, Wigström et al 1986). These studies showed that some consequence of postsynaptic depolarization is necessary to enable LTP induction at just those synapses that are eligible to change by virtue of being active at about the same time. This interactive mechanism can operate (in vitro) in the absence of sodium spikes and seems to have the required spatiotemporal specificity to account for aspects of associative LTP (Kelso et al 1986, Gustafsson et al 1987, Brown et al 1989).

There is general agreement that the induction of associative LTP is

triggered by Ca^{2+} influx into the postsynaptic cell (Eccles 1983, Bliss & Lynch 1988). Several investigators have suggested that the Ca^{2+} influx occurs through ionic channels gated by the *N*-methyl-D-aspartate (NMDA) subtype of glutamate receptor (Collingridge & Bliss 1987, Bliss & Lynch 1988, Cotman et al 1988, Brown et al 1988a,b, 1989). The key observation (Collingridge et al 1983; see also Muller et al 1988, Kauer et al 1988) was that antagonists of the NMDA receptor (such as AP5) block the induction of LTP but not its expression (Figure 3*B*). Much of the current debate has focused on the subcellular locus of the NMDA receptor-gated channels that are responsible for the postsynaptic Ca^{2+} influx, the dynamics and compartmentalization of the resulting increase in $[\text{Ca}^{2+}]_i$, and the molecular reactions that are controlled by this transient increase in $[\text{Ca}^{2+}]_i$. The development of a caged Ca^{2+} buffer that releases Ca^{2+} when stimulated by a UV light pulse (Tsien & Zucker 1986) provided another means of testing the idea that an increase in $[\text{Ca}^{2+}]_i$ is sufficient to induce LTP. Recent experiments have shown that cells containing this caged compound exhibit an LTP-like synaptic enhancement when exposed to a pulse of UV light (Malenka et al 1988).

ALTERNATIVE MODELS Several models have been proposed for the induction of LTP. Here we consider just three, beginning with one that we have been developing for the Sch/comm synapses (Brown et al 1988b, 1989, Zador et al 1990). In this model, which we denote *Model 1*, the trigger for synaptic enhancement is Ca^{2+} influx through NMDA receptor-gated channels that are located on the dendritic spine head. Model 1 maintains that (a) the peak transient increase in $[\text{Ca}^{2+}]_i$ is localized within the dendritic spine, (b) the spine amplifies the local change in $[\text{Ca}^{2+}]_i$, and (c) the relationship between the peak transient increase in $[\text{Ca}^{2+}]_i$ and the amount of LTP is nonlinear. In regard to this last point, it may be appropriate to think of some threshold level of $[\text{Ca}^{2+}]_i$ in the spine head as having a high probability of triggering molecular "switches" (Miller & Kennedy 1986) that initiate the modification process.

Model 1 maintains that for the NMDA receptor-gated channels on the spine head to allow Ca^{2+} influx, the receptor-iontophore complex must receive two local signals (Brown et al 1988b, 1989). One is a chemical signal (glutamate binding to the NMDA receptor) resulting from activity in the presynaptic terminal. The other is an electrical signal (strong depolarization of the dendritic spine head) that results from ongoing activity in the postsynaptic cell. The depolarization is needed to relieve a Mg^{2+} block of the channel (Nowak et al 1984, Mayer et al 1984, Jahr & Stevens 1987). The voltage across the spine head is controlled not only by activity in synapses located on this spine, but also by (a) the activity in other

A**B**

synapses located on other parts of the postsynaptic neuron, (b) the electrotonic structure of the postsynaptic neuron, and the relationship of this structure to the set of active synapses, and (c) the spatial distribution of various types of membrane conductances throughout this structure (see (Johnston & Brown 1984, Dingledine 1986, Brown et al 1988b).

Some predictions of Model 1 have already been confirmed. One prediction (Brown et al 1988b) was that LTP induction should be prevented when the postsynaptic cell is depolarized to +10 mV during the tetanic stimulation. The reason is that Ca^{2+} entry through NMDA channels is not expected at this membrane potential (MacDermott et al 1986, Mayer et al 1987). By contrast, Ca^{2+} entry through voltage-gated calcium channels should not be blocked at the potential. This prediction was confirmed for the Sch/comm synapses (Malenka et al 1988).

Model 2 differs in one major respect. In this model Ca^{2+} influx is through voltage-dependent calcium channels rather than NMDA receptor-gated channels (Gamble & Koch 1987). An extension of this model, developed for the mossy-fiber synapses of the CA3 region (Hopkins & Johnston 1988, Johnston et al 1988), maintains that for the voltage-dependent calcium channels to be effective, norepinephrine must act on the postsynaptic cell. This extension is consistent with the observation that LTP induction in the mossy-fiber system is blocked by β -receptor antagonists (Hopkins & Johnston 1988, Johnston et al 1988).

Model 3 also differs in one major respect. In this model Ca^{2+} influx is through NMDA receptor-gated channels located on the dendritic shaft rather than the spine head (Bliss & Lynch 1988, Wickens 1988). This model was suggested for the type of LTP that occurs in the perforant pathway input to the dentate gyrus and possibly also the Sch/comm input to the CA1 region. The rationale underlying this hypothesis and its implications are considered at length elsewhere (Bliss & Lynch 1988, Brown et al 1989, Wickens 1988).

Figure 3 Role of NMDA receptors in the induction but not the expression of LTP. *A.* Schematic of the hippocampal brain slice in relationship to stimulating and recording electrode (for additional detail, see Brown & Zador 1990). A cell in the CA1 region was impaled with an intracellular electrode, while field potentials were monitored with an extracellular electrode. Two sets of synaptic inputs were stimulated independently with two separate electrodes (S1 and S2). *B. Upper panel:* Bath application of 100 μM DL-AP5 did not attenuate the S1-produced EPSPs, but did prevent the induction of LTP following either tetanic stimulation (three trains 12 seconds apart, 100 Hz for 200 msec each) alone (*open arrow*) or when paired (*open and solid arrows*) with simultaneous outward current steps (1.5 nA, 200 msec). *Lower panel:* Addition of DL-AP5 did not block the expression of LTP in the S2 pathway that was induced in medium prior to the addition of DL-AP5. Modified from Brown & Zador (1990).

Simulations of the synaptic activity-dependent $[Ca^{2+}]_i$ dynamics have only been performed for Model 1 (Zador et al 1990) and Model 2 (Gamble & Koch 1987), but the essential features of all three models can be appreciated intuitively. Cooperativity emerges naturally in all three models from the relationship between the strength of the stimulated synaptic inputs and the resulting Ca^{2+} influx. Cooperativity has not yet been demonstrated to be a feature of LTP induction at the mossy-fiber synapses. If the predicted cooperativity cannot be demonstrated at the mossy-fiber synapses, Model 2 will have to be rejected or substantially revised.

A related prediction of Model 2 is that the induction of LTP in the mossy-fiber synapses should be controllable solely by manipulations of the postsynaptic membrane potential. Specifically, LTP induction should be prevented if the postsynaptic membrane potential is maintained under voltage-clamp conditions at a relatively negative potential during tetanic stimulation of the mossy-fiber synaptic inputs. One version (Gamble & Koch 1987) of this model implies that it should be possible to induce LTP experimentally simply by strongly depolarizing the postsynaptic neuron in the absence of presynaptic stimulation. This prediction results because Ca^{2+} influx into the spines is not directly dependent on transmitter release but only on the potential at the spine head, which should be controllable from the soma (Brown et al 1988b). Neither prediction has been adequately assessed in the mossy-fiber synapses.

The three models also differ in regard to the occurrence of associative (Figure 1A) versus heterosynaptic (Figure 1B) LTP. Model 1 contains strong constraints against heterosynaptic LTP (Brown et al 1988b, 1989, Zador et al 1990). Model 2 predicts that heterosynaptic should occur among mossy-fiber synapses. Because the mossy-fiber synapses are electrotonically near each other (Brown et al 1988b), tetanic stimulation of an *S* input should cause sufficient depolarization in spines associated with unstimulated synaptic inputs to induce LTP in them. In the absence of further constraints, Model 3 might also predict heterosynaptic LTP among closely adjacent synapses (see Brown et al 1989). Additional details and other differential and testable predictions of the models are reviewed at length elsewhere (Brown et al 1988b, 1989).

SPATIOTEMPORAL SPECIFICITY The spatiotemporal specificity of associative LTP is easy to appreciate in Model 1. The spatial (or input) specificity results from the fact that Ca^{2+} influx into the spine head requires glutamate binding to NMDA receptors on the subsynaptic membrane. The physical geometry of the spine, plus Ca^{2+} buffering systems, restrict the large, synaptically produced, transient changes in $[Ca^{2+}]_i$ to the spine region

(Zador et al 1990). Therefore only active synapses can undergo enhancement. This input specificity is not a requirement of Model 2. As indicated above, the latter model is consistent with the occurrence of heterosynaptic LTP (see Figure 1*B*). Without further constraints, input specificity is also not achieved in Model 3.

In Model 1, the temporal specificity arises from the fact that the NMDA receptor-iontophore complex must receive two signals simultaneously for the iontophore to become highly permeable to Ca^{2+} . Glutamate must be bound to the NMDA receptor, and the membrane containing the iontophore must be sufficiently depolarized to relieve the Mg^{2+} block. Thus tetanic stimulation of a *W* input fails to cause synaptic enhancement (Figure 1*A*, panel 1) unless this stimulation is accompanied by sufficiently strong postsynaptic depolarization (Figure 1*A*, panel 3). Note that the model does not require that presynaptic stimulation and postsynaptic depolarization occur simultaneously (Zador et al 1990). The maximum amount of glutamate binding occurs with a short delay after each presynaptic stimulation. Glutamate unbinding from the NMDA receptor takes additional time. After unbinding, glutamate either binds again, diffuses out of the synaptic cleft, or becomes sequestered intracellularly. Because complete removal of glutamate from the synaptic cleft after each presynaptic stimulation takes time, LTP could be induced in principle even if there were a small temporal gap—a brief “trace period”—between the end of a *W* presynaptic stimulation and the subsequent onset of a strong postsynaptic depolarization (Brown et al 1989). Simulations of Model 1 predict a trace period of up to tens of milliseconds (Zador et al 1990).

Two groups of investigators (Levy & Steward 1983, Gustafsson et al 1987, Levy & Burger 1987) have in fact reported the occurrence of LTP even after a trace period of 10 msec or longer (reviewed in Brown et al 1989). There is some basis for postulating that the removal of glutamate from the synaptic cleft is relatively slow. One reason is that the component of postsynaptic current associated with NMDA receptor activation, sometimes called the NMDA current, appears to be relatively persistent in hippocampal neurons (for review see Mayer & Westbrook 1987; see also Forsythe & Westbrook 1988). The existence of what we have called a “forward-pairing” trace period (Brown et al 1989) could be relevant to some of the computations that might be carried out in a network of such synapses (Tesauro 1986).

Mechanism of LTP Expression and Maintenance

The mechanisms responsible for the expression and maintenance of LTP are less well understood than those described above for the induction

process. The important questions concern the locus of change and its nature. Results of recent pharmacological studies have suggested that, although LTP *induction* depends on NMDA receptor activation (Collingridge et al 1983, Kauer et al 1988, Muller et al 1988), the *expression* of LTP depends on non-NMDA receptors (Kauer et al 1988, Muller & Lynch 1988, Muller et al 1988) (see Figure 3*B*). Some of these results have been interpreted as demonstrating a postsynaptic locus for the biochemical changes that underlie LTP expression and maintenance. The published results are indeed suggestive, but they are not yet sufficient to rule out all possible presynaptic mechanisms.

Neuromodulation of LTP Induction and Maintenance

There may be multiple controls—both local and extrinsic—over the extent and nature of most if not all use-dependent synaptic modifications. All of the preceding LTP induction models can easily accommodate an external signal that controls the overall plasticity of a brain region. Such signals might set the occasion for learning. Any condition that reduces γ -aminobutyric (GABA)-mediated inhibition (and therefore increases postsynaptic depolarization) would be expected to increase the magnitude of LTP or the probability of its occurrence. Pharmacological disinhibition is known (in vitro) to have this effect in the hippocampus (Wigström & Gustafsson 1983, Pacelli et al 1987).

Disinhibition is not the only route through which a neuromodulator might affect use-dependent plasticity. Johnston and co-workers have shown that pharmacological agonists and antagonists of the noradrenergic β -receptor can control the magnitude, probability of occurrence, or persistence of LTP in the mossy-fiber synapses (Hopkins & Johnston 1988, Johnston et al 1988). Although endogenously secreted norepinephrine would appear to lack the spatial specificity to instruct change selectively at any particular synapse, in principle it could serve as an extrinsic “global teacher” signal (Brown et al 1988b) that enables or preserves changes in groups of active synapses. In this way, a neuromodulatory signal can interact with a Hebbian mechanism to produce a foundation for “reinforcement learning.” To understand selective information storage in the nervous system, we need to learn more about neuromodulatory control of the induction and persistence of use-dependent synaptic modifications. After LTP has been induced, can its persistence be controlled by a subsequent neuromodulatory action? Is norepinephrine or some other neuromodulator a signal for the “consolidation” (Livingston 1967b, Kety 1972, Gold & McGaugh 1975, McGaugh 1988, 1989) of use-dependent synaptic modifications?

GENERALIZED HEBBIAN SYNAPTIC MECHANISMS

Theoretical studies and common sense point to the utility of being able both to increase and to decrease synaptic strength. In principle, such synapses can be constructed by combining an interactive form of synaptic enhancement with either an interactive or a noninteractive form of synaptic depression (Palm 1984, Linsker 1986, 1988a,b, Kairiss et al 1988). The question is whether these different forms of use-dependent synaptic plasticity in fact co-occur at the same synapses. We know from research on *Aplysia* that use-dependent forms of enhancement and depression can combine at the same synapses in interesting ways that allow learning-related adaptive modifications to occur (Hawkins & Kandel 1984, Carew & Sahley 1986). However, the synaptic enhancement that has been demonstrated thus far in *Aplysia* is not Hebbian (Carew et al 1984). Below we first consider the possibility that an interactive mechanism for enhancement may co-occur at synapses that exhibit an interactive or noninteractive mechanism for depression.

Associative LTP and Heterosynaptic Depression in Dentate Gyrus

Levy & Burger (1987) have suggested that the synapses of the perforant pathway inputs to the dentate gyrus undergo modifications that conform to what we have termed a generalized Hebb rule. Both associative LTP (Figure 1A) and heterosynaptic depression (Figure 1C) have been reported in these synapses (Levy & Steward 1979, 1983, Levy 1985, Levy & Desmond 1985). In abstracting their experimental results, Levy & Burger (1987) assumed an interactive form of heterosynaptic depression, which was embedded in a single modification rule of the general type given by Eq. 6 (Figure 4, discussed below). In this equation the level of presynaptic activity determines the direction of synaptic change. Synaptic depression occurs if the presynaptic activity level is low when the postsynaptic activity level is high. Synaptic enhancement occurs if presynaptic activity is high when postsynaptic activity is high. The amount of synaptic change is linearly proportional to the level of postsynaptic activity. Associative LTP induction in this system is known to involve the NMDA receptor (Collingridge & Bliss 1987). Two interesting and unresolved questions are whether the induction of heterosynaptic depression in the perforant pathway depends on NMDA receptor activation and whether it involves an interactive mechanism.

Associative LTP and Homosynaptic Depression in Hippocampus

Stanton & Sejnowski (1989) looked for a covariance-type of rule (Sejnowski 1977a,b) in the Sch/comm synaptic input to the CA1 pyramidal

neurons. In agreement with previous results, they found that the synapses display associative LTP when pre- and postsynaptic activity are positively correlated or active "in phase." The new finding was that when pre- and postsynaptic activities were negatively correlated or active "out of phase," a long-term depression was produced. A similar depression was reported when the inputs were activated while the postsynaptic membrane was hyperpolarized. The effect seemed to be long-lasting and restricted to the activated synapses. It did not depend on activation of NMDA receptors. This homosynaptic depression appears to be different from the heterosynaptic depression reported in the dentate gyrus. These results are suggestive of the type of covariance mechanism that is discussed further below.

Synaptic Plasticity in Developing Visual Cortex

Numerous investigators (Stent 1973, Shatz & Stryker 1978, Bienenstock et al 1982, Linsker 1986, Bear & Cooper 1989, Pearson et al 1987, Reiter & Stryker 1988, Constantine-Paton et al 1990) have proposed various extensions or generalizations of the basic Hebb rule to explain ocular dominance plasticity and other experience-dependent aspects of visual cortical development. Ocular dominance plasticity refers to an experience-dependent shift in the relative responsiveness of visual cortical neurons to input from the two eyes. The classical observation was that when vision in one eye is occluded during a critical period of development, an ocular dominance shift occurs in favor of the nonoccluded eye (Wiesel & Hubel 1963, 1965). Stent (1973) interpreted these results in terms of a synaptic modification rule that combined a simple Hebb rule with a noninteractive form of synaptic depression. Because of the interactive enhancement, postsynaptic activity driven by the nonoccluded eye strengthened all co-active synapses, which would be those from the nonoccluded eye. Because of the noninteractive depression, this same postsynaptic activity weakened all synapses that were not co-active, which would be those from the occluded eye.

The two components of Stent's generalized Hebbian modification are similar to certain of the synaptic modifications considered above. An interactive enhancement of the sort that Stent required could be implemented by the type of NMDA receptor-dependent LTP mechanism that we previously discussed in regard to the Sch/comm synapses. If the distribution of NMDA receptors is a guide to the occurrence of Hebbian synapses, we would expect to find such synapses in several regions of the neocortex, including the visual cortex (Monaghan & Cotman 1985, Cotman et al 1988). Although much less is known about the biophysics of synaptic transmission in neocortex than in the hippocampus, evidence for

a Hebbian form of LTP is growing. Baranyi & Szenté (1987) reported (in vivo) the involvement of postsynaptic activity in the induction of LTP in some cells of cat motor cortex (see also Baranyi & Fehér 1981). Additionally, Bindman et al (1988) discovered (in vitro) a Hebb-like mechanism in rat sensorimotor cortex—one that shows a spatial specificity similar to that found in hippocampus. Furthermore, an NMDA receptor-dependent form of LTP has recently been reported in visual cortex (Artola & Singer 1987, Connors & Bear 1988, Kimura et al 1988). Finally, pharmacological blockade of NMDA receptors has been shown to prevent certain forms of experience-dependent plasticity in visual cortex (Singer et al 1986, Bear et al 1987, Kleinschmidt et al 1987, Rauschecker & Hahn 1987; see also Cline et al 1987). The noninteractive synaptic depression could be implemented by some form of heterosynaptic depression (Figure 1C), which is unfortunately still poorly understood.

Reiter & Stryker (1988) recently discovered a very different effect of monocular occlusion when the visual cortex is pharmacologically inhibited by local application of muscimol (an agonist for the GABA-A receptor). They reported that under these conditions, ocular dominance shifted in favor of the less active synapses from the occluded eye. This finding led them to conclude that the level of postsynaptic activity controls the direction of cortical plasticity. Bear et al (1987) previously reached a similar conclusion. Reiter & Stryker (1988) explained this interaction between muscimol and monocular occlusion on ocular dominance plasticity as follows:

This suggests the following type of “learning rule” for ocular dominance plasticity: the connections between the more-active open eye inputs and strongly inhibited postsynaptic cells are weakened relative to the inputs from the less-active closed eye, while the reverse happens when the postsynaptic cells are not inhibited completely. This learning rule suggests the presence of a threshold membrane potential that determines the direction of an ocular dominance shift.

To account for all of the above ocular dominance shifts in terms of known synaptic mechanisms, one could postulate the co-occurrence of three forms of plasticity in the visual cortex (cf. Kairiss et al 1988): an interactive synaptic enhancement, a noninteractive heterosynaptic depression, and a noninteractive homosynaptic depression (see Figure 1 and Eq. 12). Alternatively, the combination of an interactive form of synaptic enhancement and an interactive form of synaptic depression can also be made to fit the results.

Interactive Mechanisms for Synaptic Depression

It is natural to wonder whether there are interactive mechanisms for synaptic depression. There are, of course, numerous examples of homo-

synaptic depression (Figure 1D) (Bruner & Kennedy 1970, Zucker 1972, Castelluci & Kandel 1974, Kandel 1976, Krasne 1976, 1978, Zucker & Bruner 1977, Rinaldi et al 1987) and heterosynaptic depression (Figure 1C) (Levy & Steward 1983, Levy & Burger 1987, Montorolo et al 1988), but these have not been explicitly shown to involve an interactive mechanism. Very recent work raises the possibility that some synapses do have an interactive mechanism for depression.

One line of evidence comes from studies of visual cortical neurons. Frégnac and co-workers have shown that an iontophoretically produced decrease in postsynaptic activity, when coupled with presynaptic activation, causes a long-lasting decrease in the response to activation of these same synapses (Frégnac et al 1988, Frégnac & Shulz 1989). Conversely, these neurons increased their response to stimuli that were paired with an iontophoretically induced increase in postsynaptic activity. Another candidate possibility is raised by the work of Stanton & Sejnowski (1989) mentioned above. The experimental tools now exist for determining whether the depression that they reported, which occurs when presynaptic activity is paired with postsynaptic hyperpolarization, involves an interactive mechanism. A third synaptic system that may be controlled by an interactive form of depression is the parallel fiber input to the Purkinje cells of the cerebellum (Ekerot & Kano 1985, Crepel & Krupa 1988, Ito 1989).

Some interesting speculations have been made about the biochemical mechanisms underlying interactive forms of synaptic depression. One proposal (Bear & Cooper 1989, in press) is that decreases in synaptic strength are mediated via second messenger systems, such as inositol triphosphate (IP_3) or diacyl glycerol (DG), that are activated by non-NMDA receptors, whereas increases in synaptic strength result from calcium influx through NMDA receptor-activated channels.

ABSTRACT REPRESENTATIONS OF HEBBIAN MODIFICATIONS

Levy & Burger (1987) estimate that perhaps as many as 50–100 different algebraic equations have been used to describe theoretical synaptic activity-modification relationships or learning rules. Here we select just a few of these rules or algorithms that satisfy our definition of a Hebbian modification. The selection was further limited to simple algorithms that have or could be argued to capture abstractly some important feature of known or suspected neurophysiology. The shortcomings of these abstractions can be even more informative than any claimed successes. In addition, they provide a convenient, higher-level language for communicating and

exploring theoretical relationships among changes at the synaptic, network, systems, and behavioral levels (Donegan et al 1989).

Obviously these Hebbian algorithms are not intended as substitutes for detailed and realistic biophysical models of synaptic function (cf. Zador et al 1990). But those interested in developmental and cognitive aspects of computational neuroscience can reasonably argue that some extreme simplifications will inevitably be required to appreciate self-organization and learning in the mammalian brain (see Kohonen 1984, 1987, Linsker 1986, Arbib 1987, Churchland et al 1990, Sejnowski & Churchland 1989, Sejnowski et al 1988). The challenge at every level of organization is to determine which facts are theoretically important and which are not (see Crick & Asanuma 1986, Sejnowski & Churchland 1989).

Hebbian Algorithms

Although Hebb did not attempt to quantify the relationship between post- and presynaptic firing and the ensuing synaptic enhancement, his idea is commonly translated into discrete-time equations of the form:

$$w_{ij}(t+1) = w_{ij}(t) + \Delta w_{ij}(t), \quad 1.$$

where

$$\Delta w_{ij}(t) = F[a_i(t), a_j(t)], \quad 2.$$

t is discrete time, w_{ij} is the "weight" of the connection from presynaptic unit u_j to postsynaptic unit u_i , $a_j(t)$ and $a_i(t)$ are some measures of pre- and postsynaptic activity, and the change in synaptic efficacy $\Delta w_{ij}(t)$ is some function¹ F of both pre- and postsynaptic activity.

The synaptic weight w_{ij} is assumed to be a suitable abstract representation of "synaptic strength." The relationship between w_{ij} and the variables in contemporary biophysical models of synaptic transmission is usually unspecified. Commonly w_{ij} is treated as a dimensionless variable. For concreteness we assume here that w_{ij} is proportional to the expected value of a multiparameter quantal probability model (cf. Brown et al 1989) of quantal synaptic transmission. The activity terms $a_j(t)$ and $a_i(t)$ are also often treated as dimensionless variables. Here they are assumed to be proportional to some measure of a biophysical variable such as the spike frequency during a short time interval, the potential across a designated patch of membrane, the conductance to Ca^{2+} or another ion in some membrane region, the current carried by Ca^{2+} or another ion through a

¹ F is most appropriately termed a functional because the activity terms themselves are functions of time.

given area of membrane, or the intracellular activity of Ca^{2+} or some other second messenger in a particular subcellular compartment.

SPIKE PRODUCT RULE Much of what follows concerns the form of the functional F in Eq. 2. Perhaps the simplest functional is a product of the pre- and postsynaptic spike frequencies:

$$w_{ij}(t+1) = w_{ij}(t) + c\bar{x}_i(t)\bar{x}_j(t), \quad 3.$$

where $\bar{x}_i$ and $\bar{x}_j$ are the mean firing frequencies of the post- and presynaptic units, respectively (averaged over the previous fraction of a second), and c is a proportionality constant that determines the learning rate (cf. Levy & Desmond 1985, Klopff 1988, 1989, Sejnowski & Tesauro 1989). The three key assumptions in Eq. 3 are that the contributions of pre- and postsynaptic activity can be separated into two corresponding activity terms, that the proper measurement of activity is mean spike frequency, and that the nature of the interaction between pre- and postsynaptic spiking frequency is a simple product.

Our current understanding of associative LTP is not well-represented by these assumptions. The causally relevant postsynaptic variable may be depolarization of the spine head rather than spike frequency (Kelso et al 1986, Brown et al 1988b, 1989, Zador et al 1990). The choice of presynaptic spike frequency for the presynaptic term may also be inappropriate. Although we commonly use high-frequency presynaptic stimulation to induce LTP (Brown et al 1989), this use is not necessary. LTP can also be induced by pairing single presynaptic stimulations with properly timed postsynaptic depolarizations (Wigström et al 1986). These pairings do not have to occur at a high repetition rate to be effective. The relative efficacy of different repetition rates is still unclear. Several groups have shown that activity patterns at approximately the theta frequency (5–7 Hz) are particularly effective (Larson & Lynch 1986, Larson et al 1986, Rose & Dunwiddie 1986, Staubli & Lynch 1987, Greenstein et al 1988), probably because of disinhibition at this frequency (Pacelli et al 1987).

ACTIVITY PRODUCT RULE We can generalize Eq. 3 by not specifying the activity measurement:

$$w_{ij}(t+1) = w_{ij}(t) + g[a_i(t)]h[a_j(t)]c, \quad 4.$$

where $g[a_i(t)]$ reflects some measure of postsynaptic activity $a_i(t)$, and $h[a_j(t)]$ indicates some measure of presynaptic activity $a_j(t)$, both taken at the same time t . Equation 4 is an improvement over Eq. 3 because it leaves open the proper specification of the pre- and postsynaptic activity functions. Nevertheless, this general form of synaptic learning rule (Figure

4, *left curve*) still leaves much to be desired, not only as an abstraction of known synaptic physiology but also from a psychological and computational perspective.

The first and most obvious problem is that Eq. 4 requires successive pairings of pre- and postsynaptic activity to cause equal increments in synaptic strength. Unlike any known synaptic modification, w_{ij} can therefore increase without bound. The LTP mechanism saturates after one or just a few stimulations (deJonge & Racine 1985), in sharp contrast to the behavior of Eq. 4. Clearly, networks of synapses governed by Eq. 4 might not lend themselves easily to the negatively accelerated acquisition curves of the sort captured by the learning equation of Rescorla & Wagner (1972):

$$\text{Hebb rule: } \Delta w_{ij} = g(a_i)[h(a_j)]c$$

$$\text{Generalized Hebb rule: } \Delta w_{ij} = g(a_i)\{[h(a_j)]c - w_{ij}\}$$

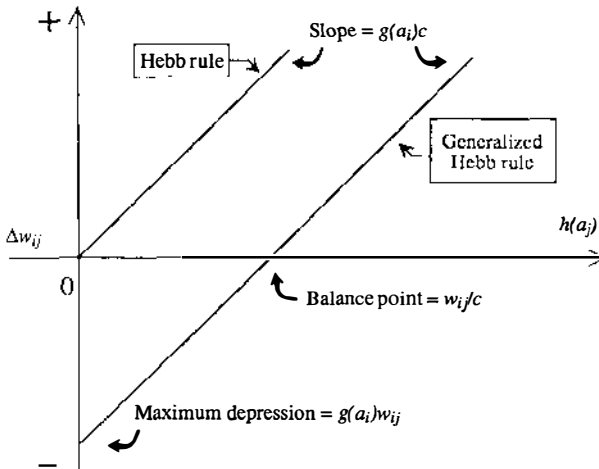


Figure 4 Abstract representation of Hebbian modifications. Ongoing presynaptic activity in the axonal bouton is represented by $h(a_j)$; ongoing postsynaptic activity is represented by $g(a_i)$; and the connection weight is represented by w_{ij} . In the Hebbian activity product rule (Eq. 4, plot on *left*) the change in weight Δw_{ij} is the product of the presynaptic activity function $h(a_j)$, the postsynaptic activity function $g(a_i)$, and a proportionality constant c that determines the rate of change. The Hebb rule only allows positive values of Δw_{ij} , the magnitude of which is proportional to the slope $g(a_i)c$. In one type of generalized Hebb activity product rule (Eq. 6, plot on *right*), the presynaptic activity function is replaced with $\{[h(a_j)]c - w_{ij}\}$. This rule allows positive or negative values of Δw_{ij} depending on whether $h(a_j)$ exceeds or is less than the balance point w_{ij}/c . The magnitude of Δw_{ij} is again proportional to $g(a_i)c$.

$$\Delta V_n = K(\lambda - V_{n-1}), \quad 5.$$

where ΔV_n is the change in associative strength on trial n , K is a learning rate parameter reflecting stimulus salience, and λ is the asymptotic associative strength.

The second problem is that unless ongoing pre- and postsynaptic activities are perfectly negatively correlated, w_{ij} will always tend to increase due to chance coincidences (Grossberg 1976, Sejnowski 1981, Kohonen 1984). Thus w_{ij} will automatically tend to grow without bound and its value can never decrease. This in turn leads to "runaway instability" (Sejnowski & Tesauro 1989). Once the w_{ij} associated with some presynaptic unit u_i is large enough, subsequent activity in u_i will always increase w_{ij} regardless of coactivity in other presynaptic units. One obvious solution is simply to clip Eq. 4 at some upper limit on w_{ij} (Tesauro 1986). In the absence of passive decay, however, clipping would introduce the problem of saturation. Spurious coincidences plus clipping will cause a uniform set of w_{ij} to evolve to some asymptotic level automatically. As the set of w_{ij} approaches its maximum value, all information represented in the synaptic weights is of course lost. New memories could not be stored and old ones would be obliterated.

Some of these problems can be avoided by adding a passive decay term to Eq. 4. Thus we might prefer to assume that in the absence of synaptic activity, the current w_{ij} is not necessarily permanent. Instead w_{ij} can relax back to other states as a function of the immediate and long-term history of the synapse. Such a learning rule could be useful for temporary information storage, but this solution reintroduces the problem of how to achieve more persistent representations. In this context we should note the lack of experimental evidence that LTP is permanent. In fact, the best information suggests that experimentally induced LTP decays over the course of hours, days, or weeks (Racine et al 1983, Reymann et al 1985). Thus hippocampal LTP might be involved in some form of temporary information storage.

Generalized Hebbian Algorithms

In the previous section we considered some problems with simple product rules that permit only increases in w_{ij} . Stable network function may sometimes require modification algorithms that cause both increases and decreases in w_{ij} . Here we consider various extensions of Hebb algorithms that allow both to occur. We refer to these as generalized Hebbian algorithms.

RENORMALIZATION Von der Malsburg (1973) proposed that a "renormalization" process occurs each time a w_{ij} increases. Renormalization forces the combined synaptic weights associated with each postsynaptic

unit (Σw_{ij} for each u_i) to remain constant. This zero-sum constraint on Δw_{ij} results in both increases and decreases in the connection strengths. At a descriptive level, one can think of renormalization in terms of presynaptic "competition" for connection strength (see also Volper & Hampson 1987), an idea that is sometimes invoked to explain aspects of neural development (Artola & Singer 1987, Constantine-Paton et al 1990) and damage-induced plasticity (Merzenich et al 1984).

ACTIVITY PRODUCT RULE Product rules have been modified from Eqs. 3 and 4 in various ways to allow for reversible modifications in synaptic strength. One approach is to substitute for the presynaptic term in Eq. 3 an expression such as $(x_j - x_b)$, where x_b is the average background activity in the afferent input (Kohonen 1984). Another approach (cf. Levy & Desmond 1985, Levy & Burger 1987, Kohonen 1988) is to replace the presynaptic term in Eq. 4 with an expression such as $\{h[a_j(t)]c - w_{ij}(t)\}$, where c is a learning rate constant. We can write the generalized Hebb algorithm as follows:

$$w_{ij}(t+1) = w_{ij}(t) + g[a_i(t)] \{h[a_j(t)]c - w_{ij}(t)\}, \quad 6.$$

where the postsynaptic term $g[a_i(t)]$ can represent the activity in a particular spine on neuron i at time t . Equation 6 implies that for afferent inputs in which $h[a_j(t)]c < w_{ij}(t)$, the synaptic strength will decrease by an amount proportional to the postsynaptic activity $g[a_i(t)]$ (Figure 4, *right curve*). The synaptic strength increases in proportion to $g[a_i(t)]$ when $h[a_j(t)]c > w_{ij}(t)$. Thus the activity "balance point" (see Figure 3) for modifying w_{ij} is a variable (w_{ij}/c) that is proportional to the value of w_{ij} at the time of presynaptic activation. This approach eliminates the problem of runaway instability and results in a negatively accelerated synaptic modification curve.

We suggested above that the type of modification expressed by Eq. 6 could be relevant to Stent's (1973) account of the classical experiments of Wiesel & Hubel (1963, 1965). Other results (Reiter & Stryker 1988), however, could not be explained by a modification rule with a presynaptic balance point. In their studies of visual cortical development, Cooper and co-workers considered an algorithm in which the balance point for synaptic modification was associated with the postsynaptic term (Bienenstock et al 1982, Bear et al 1987). Their algorithm, which is mathematically more complicated than Eq. 6, holds that the modification threshold changes nonlinearly with the time-averaged postsynaptic activity. Low activity levels cause a decrease in the threshold, effectively enabling weak inputs to increase their strength. High levels drive the threshold up. This prevents saturation and stabilizes the network in a state of maximum selectivity.

Later we consider algorithms that have both pre- and postsynaptic balance points.

ACTIVITY TRACE PRODUCT RULE Recall that LTP can be induced even with a brief time interval between the end of a presynaptic stimulation and the subsequent onset of a sufficiently strong postsynaptic depolarization (Levy & Burger 1987). Similarly, in certain variations or extensions of a Hebb-like algorithm, the product of earlier presynaptic activity (at time $t - \tau$) and later postsynaptic activity (at time t) is what generates the changes (Tesauro 1986, Klopff 1988, Sejnowski & Tesauro 1989). To allow for a "forward-pairing trace period" (cf. Levy & Burger 1987, Brown et al 1989), the presynaptic term in Eq. 6 can be offset by time τ . The differential effects of several values of τ can be included by modifying Eq. 6 as follows (cf. Klopff 1988, 1989):

$$w_{ij}(t+1) = w_{ij}(t) + g[a_i(t)] \sum_{\tau=0}^k \{h[a_j(t-\tau)]c_\tau - w_{ij}(t-\tau)\}, \quad 7.$$

where c_τ is a positive coefficient that determines the contribution of each offset.

ACTIVITY COVARIANCE RULE Sejnowski (1977a,b) proposed a modification rule in which changes in synaptic strength are proportional to the covariance between pre- and postsynaptic spiking (see Eqs. 7–15 of Sejnowski 1981). If we substitute activity for spiking, a discrete-time version of Sejnowski's learning algorithm can be written as follows:

$$w_{ij}(t+1) = w_{ij}(t) + \text{cov}[a_j(t), a_i(t)], \quad 8.$$

where

$$\text{cov}[a_j(t), a_i(t)] = \gamma[a_j(t)a_i(t) - \bar{a}_j(t)\bar{a}_i(t)], \quad 9.$$

γ is a proportionality constant that determines the learning rate, and $\bar{a}_j(t)$ and $\bar{a}_i(t)$ are the expected values of the activities at time t or the average values over some previous time interval. According to this scheme, w_{ij} increases if pre- and postsynaptic activities are positively correlated and decreases if they are negatively correlated. Uncorrelated activity has no effect on w_{ij} . Equation 8 solves the problem of automatic growth owing to spurious coincidences.

Linsker (1986) has used a covariance-type rule to simulate aspects of experience-dependent self-organization of the visual system. Extending this work, Linsker (1988a,b) has proposed an organizing principle for understanding simple layered networks in terms of Shannon information theory. He has shown that Hebb rules can give rise to units with the

property of *maximum information preservation* in which each unit maximizes the information that the output signal conveys about the input signal, subject to constraints and noise. By maximizing the Shannon information rate of its output, each unit essentially extracts and summarizes the most relevant features of its input. Linsker has termed this the *informatmax* principle. What this means is that, within certain limits, simple layered networks comprised of Hebbian connections can learn in an “unsupervised” fashion to perform a principle-components analysis on the incoming information. This ability to self-organize without a teacher seems well-suited to experience-dependent or experience-expectant aspects of perceptual development.

FURTHER GENERALIZATIONS We have mentioned three types of generalized Hebb algorithms. In Eq. 6 the direction of modification is determined by the level of presynaptic activity. The presynaptic balance point in Eq. 6 is illustrated graphically in Figure 4 (*right curve*). In the type of rule proposed by Cooper’s group (Bear et al 1987), the direction of modification is controlled by the level of postsynaptic activity. There is a postsynaptic balance point. In the type of rule that Sejnowski (1977) explored, the balance point is not localized on either side of the synapse. In Sejnowski’s covariance rule, the direction of change is a joint function of both pre- and postsynaptic activity (Eq. 9). This is also true of Linsker’s (1986, 1988b) rule, a discrete-time version of which can be written as follows:

$$w_{ij}(t+1) = w_{ij}(t) + F[(a_j(t) - A_j)(a_i(t) - A_i)], \quad 10.$$

where A_j and A_i are constants. If F is a linear functional, Eq. 10 may be expanded to:

$$w_{ij}(t+1) = w_{ij}(t) + \alpha a_j(t)a_i(t) + \beta a_i(t) + \gamma a_j(t) + \delta. \quad 11.$$

This bilinear equation is identical to that discussed by Palm (1984; see also Linsker 1988b). It contains one interactive term with coefficient α , two noninteractive terms with coefficients β and γ , and a constant term δ . If α is positive and β and γ are negative, then Eq. 11 represents a generalized Hebb rule in which $\beta a_i(t)$ is a noninteractive type of heterosynaptic depression and $\gamma a_j(t)$ is a noninteractive type of homosynaptic depression.

Equation 11 can be further generalized as follows:

$$w_{ij}(t+1) = w_{ij}(t) + \alpha[a_j(t)a_i(t)] + \beta[a_i(t)] + \gamma[a_j(t)] + \delta, \quad 12.$$

where α , β , and γ are no longer simple coefficients but instead functionals that depend upon pre- and/or postsynaptic activity. The idea is the same, however. There is an interactive form of enhancement and two noninteractive forms of depression. Equation 12 has several attractive features: it

appears consistent with both types of ocular dominance plasticity effects mentioned above; it avoids the problem of automatic growth due to spurious coincidences; and it is a plausible implementation in the sense that neurobiological analogs to each of its activity terms exist (Kairiss et al 1988). Equation 12 also raises several interesting questions. Do the required three forms of plasticity co-occur in the same synaptic system? How persistent are they? What is the best mathematical representation of the activity-modification relationships? Is anything gained computationally by replacing the two noninteractive forms of depression with a single interactive form of depression? Do interactive forms of depression exist?

Global Control of Learning

Several investigators have considered the possibility of "now print" (Livingston 1967a,b, Anderson 1985, von der Malsburg & Bienenstock 1986) or "global teacher" (Brown et al 1988b) signals. Levy & Burger (1987) address this possibility at the algorithmic level by introducing a multiplicative variable P , which takes on values between 0 and 1. Adding this variable to Eq. 6 yields:

$$w_{ij}(t+1) = w_{ij}(t) + g[a_i(t)] \{h[a_j(t)]c - w_{ij}(t)\} P. \quad 13.$$

Equation 13 assumes that three multiplicative terms corresponding to presynaptic activity, postsynaptic activity, and overall plasticity can be isolated meaningfully. If there is a physiological correlate of the P term, it could be a neuromodulatory substance such as norepinephrine (cf. Bear & Singer 1986, Kasamatsu 1987) or acetylcholine (cf. Williams & Johnston 1988, Metharate et al 1987, Sillito 1987), as mentioned above.

An alternative algorithm, suggested by Alspector et al (1988), involves "excess reinforcement." The change in synaptic weight is proportional to the correlation between the activity product and a global reinforcement signal r :

$$R = \langle r a_i a_j \rangle - \langle r \rangle \langle a_i a_j \rangle, \quad 14.$$

where $\langle \rangle$ indicates an expected value or a time-averaged quantity. The value of r is set to +1 if the output is correct, and -1 if incorrect. The learning algorithm increases w_{ij} if $R > 0$, and decreases w_{ij} if $R < 0$. An electronic implementation of a neural network using this algorithm is discussed in Alspector et al (1988).

SUMMARY AND CONCLUSIONS

We have examined the evolution of the concept of a Hebbian synaptic modification and have suggested a contemporary definition. The biophysi-

cal mechanism demonstrated in vitro to control the induction of one type of hippocampal LTP has been shown to satisfy our definition of a Hebbian synaptic modification. Whether this biophysical mechanism is involved in the organization of behavior in the manner that Hebb originally envisioned remains to be seen. We have also summarized several modification algorithms that have been explored in theoretical studies of learning in adaptive networks. These algorithms also satisfied our definition of a Hebbian modification, but their relationships to known neurobiology require further exploration. By reviewing the biophysical mechanisms and formal algorithms together, we have exposed obvious similarities and differences. Such comparisons may help bridge the gap between computational theory and knowledge of the neurobiology of use-dependent synaptic change. Current models of LTP reveal that the activity-modification relationships are extremely sensitive to the biophysical/molecular details. The activity-modification relationships obviously can have a major influence on adaptive neurodynamics at the network level. As more accurate representations of the biological complexity and diversity are introduced into adaptive network simulations, we expect to gain new insights into the classes of computation that particular networks are capable of performing.

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Literature Cited

- Abrams, T. W., Kandel, E. R. 1988. Is contiguity detection in classical conditioning a system or a cellular property? Learning in *Aplysia* suggests a possible molecular site. *Trends Neurosci.* 11: 128–35
- Alspector, J., Allen, R. B., Hu, V., Satyanarayana, S. 1988. Stochastic learning networks and their electronic implementation. In *Neural Information Processing Systems*, ed. D. Z. Anderson, pp. 9–21. New York: Am. Inst. Phys.
- Anderson, J. A. 1985. What Hebb synapses build. See Levy et al 1985, pp. 153–73
- Anderson, J. A., Rosenfeld, E., eds. 1988. *Neurocomputing: Foundations of Research*, pp. 4–14. Cambridge: MIT Press
- Arbib, M. A. 1987. *Brains, Machines and Mathematics*. New York: Springer-Verlag. 2nd ed.
- Artola, A., Singer, W. 1987. Long-term potentiation and NMDA receptors in rat visual cortex. *Nature* 330: 649–52
- Baranyi, A., Fehér, O. 1981. Synaptic facilitation requires paired activation of convergent pathways in the neocortex. *Nature* 290: 413–15
- Baranyi, A., Szente, M. B. 1987. Long-lasting potentiation of synaptic transmission requires postsynaptic modifications in the neocortex. *Brain Res.* 423: 378–84
- Barrionuevo, G., Brown, T. H. 1983. Associative long-term potentiation in hippocampal slices. *Proc. Natl. Acad. Sci. USA* 80: 7347–51
- Baxter, D. A., Bittner, G. D., Brown, T. H. 1985. Quantal mechanism of long-term synaptic potentiation. *Proc. Natl. Acad. Sci. USA* 82: 5978–82

- Bear, M. F., Cooper, L. N. 1989. Molecular mechanisms for synaptic modification in the visual cortex: Interaction between theory and experiment. In *Advances in Connectionist Theory: Neurobiological Aspects*, ed. D. Rumelhart, M. Gluck. In press.
- Bear, M. F., Cooper, L. N., Ebner, F. F. 1987. A physiological basis for a theory of synapse modification. *Science* 237: 42–48.
- Bear, M. F., Singer, W. 1986. Acetylcholine, norepinephrine and the extrathalamic control of visual cortical plasticity. *Nature* 320: 172–76.
- Bienenstock, E. L., Cooper, L. N., Munro, P. W. 1982. Theory for the development of neuron selectivity: Orientation specificity and binocular interaction in visual cortex. *J. Neurosci.* 2: 32–48.
- Bindman, L., Murphy, K. P. S. J., Pockett, S. 1988. Postsynaptic control of the induction of long-term changes in efficacy of transmission at neocortical synapses in slices of rat brain. *J. Neurophysiol.* 60: 1053–65.
- Bliss, T. V. P., Gardner-Medwin, A. R. 1973. Long-lasting potentiation of synaptic transmission in the dentate area of the unanaesthetized rabbit following stimulation of the perforant path. *J. Physiol.* 232: 357–74.
- Bliss, T. V. P., Lomo, T. 1973. Long-lasting potentiation of synaptic transmission in the dentate area of the anesthetized rabbit following stimulation of the perforant path. *J. Physiol.* 232: 331–56.
- Bliss, T. V. P., Lynch, M. A. 1988. Long-term potentiation of synaptic transmission in the hippocampus: Properties and mechanisms. See Landfield & Deadwyler 1988, pp. 3–72.
- Briggs, C. A., Brown, T. H., McAfee, D. A. 1985. Neurophysiology and pharmacology of long-term potentiation in the rat sympathetic ganglion. *J. Physiol.* 359: 503–21.
- Brown, T. H., Chang, V. C., Ganong, A. H., Keenan, C. L., Kelso, S. R. 1988a. Biophysical properties of dendrites and spines that may control the induction and expression of long-term synaptic potentiation. See Landfield & Deadwyler 1988, pp. 197–260.
- Brown, T. H., Chapman, P. F., Kairiss, E. W., Keenan, C. L. 1988b. Long-term synaptic potentiation. *Science* 242: 724–28.
- Brown, T. H., Ganong, A. H., Kairiss, E. W., Keenan, C. L., Kelso, S. R. 1989. Long-term potentiation in two synaptic systems of the hippocampal brain slice. See Byrne & Berry 1989, pp. 266–306.
- Brown, T. H., McAfee, D. 1982. Long-term synaptic potentiation in superior cervical ganglion. *Science* 215: 1411–13.
- Brown, T. H., Zador, A. 1990. The hippocampus. In *The Synaptic Organization of the Brain*, ed. G. M. Shepherd. New York: Oxford. In press.
- Bruner, J., Kennedy, D. 1970. Habituation: Occurrence at a neuromuscular junction. *Science* 169: 92–94.
- Byrne, J. H. 1987. Cellular analysis of associative learning. *Physiol. Rev.* 67: 329–439.
- Byrne, J. H., Berry, W. O., eds. 1989. *Neural Models of Plasticity*. New York: Academic.
- Cajal, S. Ramón y. 1911. *Histologie du Systeme Nerveux*, Vol. 2. Paris: Maloine.
- Callaway, E. M., Soha, J. M., Van Essen, D. C. 1987. Competition favors inactive over active motor neurons during synapse elimination. *Nature* 328: 422–26.
- Callaway, E., Soha, J. M., Van Essen, D. C. 1989. Differential loss of neuromuscular connections according to activity level and spinal position of neonatal rabbit soleus motor neurons. *J. Neurosci.* 9: 1806–24.
- Carew, T. J., Hawkins, R. D., Abrams, T. W., Kandel, E. R. 1984. A test of Hebb's postulate at identified synapses which mediate classical conditioning in *Aplysia*. *J. Neurosci.* 4: 1217–24.
- Carew, T. J., Sahley, C. L. 1986. Invertebrate learning and memory: From behavior to molecules. *Annu. Rev. Neurosci.* 9: 435–87.
- Castellucci, V. F., Kandel, E. R. 1974. A quantal analysis of the synaptic depression underlying habituation of the gill-withdrawal reflex in *Aplysia*. *Proc. Natl. Acad. Sci. USA* 71: 5004–8.
- Churchland, P. S., Koch, C., Sejnowski, T. J. 1990. What is computational neuroscience? In *Computational Neuroscience*, ed. E. Schwartz. Cambridge: MIT Press. In press.
- Cline, H. T., Debski, E. A., Constantine-Paton, M. 1987. *N*-methyl-D-aspartate receptor antagonist desegregates eye-specific stripes. *Proc. Natl. Acad. Sci. USA* 84: 4342–45.
- Collingridge, G. L., Bliss, T. V. P. 1987. NMDA receptors: Their role in long-term potentiation. *Trends Neurosci.* 10: 288–93.
- Collingridge, G. L., Kehl, S. J., McLennan, H. 1983. Excitatory amino acids in synaptic transmission in the Schaffer collateral-commissural pathway of the rat hippocampus. *J. Physiol.* 334: 33–46.
- Connors, B. W., Bear, M. F. 1988. Pharmacological modulation of long term potentiation in slices of visual cortex. *Soc. Neurosci. Abstr.* 14: 744.
- Constantine-Paton, M., Cline, H. T., Debski, E. 1990. Patterned activity, syn-

- aptic convergence, and the NMDA receptor in developing visual pathways. *Annu. Rev. Neurosci.* 13: 129–54
- Cooper, L. N., Liberman, F., Oja, E. 1979. A theory for the acquisition and loss of neuron specificity in visual cortex. *Biol. Cybern.* 33: 9–28
- Cotman, C. W., Monaghan, D. T., Ganong, A. H. 1988. Excitatory amino acid neurotransmission: NMDA receptors and Hebb-type synaptic plasticity. *Annu. Rev. Neurosci.* 11: 61–80
- Crepel, F., Krupa, M. 1988. Activation of protein kinase C induces a long-term depression of glutamate sensitivity of cerebellar Purkinje cells. An in vitro study. *Brain Res.* 458: 397–401
- Crick, F. H. C., Asanuma, C. 1986. Certain aspects of the anatomy and physiology of the cerebral cortex. In *Parallel Distributed Processing: Explorations in the Microstructure of Cognition*, ed. D. L. McClelland, D. E. Rumelhart, 2: 333–71. Cambridge: MIT Press
- deJonghe, M., Racine, R. J. 1985. The effects of repeated induction of long-term potentiation in the dentate gyrus. *Brain Res.* 328: 181–85
- Denker, J. S. 1986. Neural network models of learning and adaptation. *Physica D* 22: 216–32
- Dingledine, R. 1986. NMDA receptors: What do they do? *Trends Neurosci.* 9: 47–49
- Donegan, N. H., Gluck, M. A., Thompson, R. F. 1989. Integrating behavioral and biological models of classical conditioning. In *Computational Models of Learning in Simple Neural Systems*, ed. R. D. Hawkins, G. H. Bower. New York: Academic. In press
- Donegan, N. H., Wagner, A. R. 1987. Conditioned diminution and facilitation of the UR: A sometimes opponent-process interpretation. In *Classical Conditioning*, ed. I. Gormezano, W. F. Prokasy, R. F. Thompson, pp. 339–69. Hillsdale, NJ: Erlbaum
- Eccles, J. C. 1983. Calcium in long-term potentiation as a model for memory. *Neuroscience* 10: 1071–81
- Edelman, G. M., Gall, W. E., Cowan, W. M., eds. 1987. *Synaptic Function*. New York: Wiley
- Ekerot, C.-F., Kano, M. 1985. Long-term depression of parallel fibre synapses following stimulation of climbing fibres. *Brain Res.* 342: 357–60
- Finkel, L. H., Edelman, G. M. 1987. Population rules for synapses in networks. See Edelman et al 1987, pp. 711–58
- Forsythe, I. D., Westbrook, G. L. 1988. Slow excitatory postsynaptic currents mediated by *N*-methyl-D-aspartate receptors on cultured mouse central neurones. *J. Physiol.* 396: 515–33
- Frégnac, Y., Schulz, D. 1989. Hebbian synapses in visual cortex. In *Seeing Contour and Colour*, ed. K. J. Kilikowski. Oxford: Pergamon. In press
- Frégnac, Y., Schulz, D., Thorpe, S., Bienenstock, E. 1988. A cellular analogue of visual cortical plasticity. *Nature* 333: 367–70
- Gamble, E., Koch, C. 1987. The dynamics of free calcium in dendritic spines in response to repetitive synaptic input. *Science* 236: 1311–15
- Gold, P. E., McGaugh, J. L. 1975. A single-trace, two process view of memory storage processes. In *Short-Term Memory*, ed. D. Deutsch, J. A. Deutsch, pp. 355–78. New York: Academic
- Greenstein, Y. J., Pavlides, C., Winson, J. 1988. Long-term potentiation in the dentate gyrus is preferentially induced at theta rhythm periodicity. *Brain Res.* 438: 331–34
- Grossberg, S. 1976. Adaptive pattern classification and universal recoding: I. Parallel development and coding of neural feature detectors. *Biol. Cybern.* 23: 121–34
- Gustafsson, B., Wigström, H., Abraham, W. C., Huang, Y.-Y. 1987. Long-term potentiation in the hippocampus using depolarizing current pulses as the conditioning stimulus to single volley synaptic potentials. *J. Neurosci.* 7: 774–80
- Hawkins, R. D., Kandel, E. R. 1984. Is there a cell biological alphabet for simple forms of learning? *Psychol. Rev.* 91: 375–91
- Hayek, F. A. 1952. *The Sensory Order: An Inquiry into the Foundations of Theoretical Psychology*. Chicago: Univ. Chicago Press
- Hebb, D. O. 1949. *The Organization of Behavior*. New York: Wiley
- Hopfield, J. J., Feinstein, D. I., Palmer, R. G. 1983. “Unlearning” has a stabilizing effect in collective memories. *Nature* 304: 158–59
- Hopfield, J. J., Tank, D. W. 1986. Computing with neural circuits: A model. *Science* 233: 625–33
- Hopkins, W. F., Johnston, D. 1988. Noradrenergic enhancement of long-term potentiation at mossy fiber synapses in the hippocampus. *J. Neurophysiol.* 59: 667–87
- Ito, M. 1989. Long-term depression. *Annu. Rev. Neurosci.* 12: 85–102
- Jahr, C. E., Stevens, C. F. 1987. Glutamate activates multiple single channel conductances in hippocampal neurons. *Nature* 325: 522–25
- James, W. 1890. *Psychology: Briefer Course*. Cambridge: Harvard Univ. Press
- Johnston, D., Brown, T. H. 1984. Biophysics and microphysiology of synaptic trans-

- mission in hippocampus. In *Brain Slices*, ed. R. Dingledine, pp. 51–86. New York: Plenum
- Johnston, D., Hopkins, W. F., Gray, R. 1988. Noradrenergic enhancement of long-term synaptic enhancement. See Landfield & Deadwyler 1988, pp. 355–76
- Johnston, D., Hopkins, W. F., Gray, R. 1989. The role of norepinephrine in long-term potentiation at mossy fibers synapses in the hippocampus. See Byrne & Berry 1989, pp. 307–28
- Kairiss, E. W., Keenan, C. L., Brown, T. H. 1988. A biologically plausible implementation of a Hebbian covariance algorithm. *Soc. Neurosci. Abstr.* 14: 832
- Kandel, E. R. 1976. *The Cellular Basis of Behavior*. San Francisco: Freeman
- Kandel, E. R., Klein, M., Hochner, B., Shuster, M., Siegelbaum, S. A., et al. 1987. Synaptic modulation and learning: New insights into synaptic transmission from the study of behavior. See Edelman et al 1987, pp. 471–518
- Kasamatsu, T. 1987. Norepinephrine hypothesis for visual and cortical plasticity: Thesis, antithesis, and recent development. *Curr. Top. Dev. Biol.* 21: 367–89
- Kauer, J., Malenka, R. C., Nicoll, R. A. 1988. A persistent postsynaptic modification mediates long-term potentiation in the hippocampus. *Neuron* 1: 911–17
- Kelso, S. R., Brown, T. H. 1986. Differential conditioning of associative synaptic enhancement in hippocampal brain slices. *Science* 232: 85–87
- Kelso, S. R., Ganong, A. H., Brown, T. H. 1986. Hebbian synapses in hippocampus. *Proc. Natl. Acad. Sci. USA* 83: 5326–30
- Kety, S. 1972. Brain catecholamines, affective states and memory. In *The Chemistry of Mood, Motivation, and Memory*, ed. J. L. McGaugh, pp. 65–80. New York: Raven
- Kimura, F., Tsumoto, T., Nishigori, A., Shirokawa, T. 1988. Long-term synaptic potentiation and NMDA receptors in the rat pup visual cortex. *Soc. Neurosci. Abstr.* 14: 188
- Kleinschmidt, A., Bear, M. F., Singer, W. 1987. Blockade of “NMDA” receptors disrupts experience-dependent plasticity of kitten striate cortex. *Science* 238: 355–58
- Klopf, A. H. 1988. A neuronal model of classical conditioning. *Psychobiology* 16: 85–125
- Klopf, A. H. 1989. Classical conditioning phenomena predicted by a drive-reinforcement model of neuronal function. See Byrne & Berry 1989, pp. 104–32
- Kohonen, T. 1984. *Self-Organization and Associative Memory*. Berlin/Heldelberg: Springer-Verlag
- Kohonen, T. 1987. *Content-Addressable Memories*. New York: Springer-Verlag. 2nd ed.
- Kohonen, T. 1988. The “neural” phonetic typewriter. *Computer* 21: 11–22
- Kohonen, T., Lehtio, P., Rovano, J. 1974. Modelling of neural associative memory. *Ann. Acad. Sci. Fenn. Ser. A5* 167: 1–18
- Kohonen, T., Oja, E. 1976. Fast adaptive formation of orthogonalizing filters and associative memory in recurrent networks of neuron-like elements. *Biol. Cybern.* 21: 85–95
- Konorski, J. 1948. *Conditioned Reflexes and Neuron Organization*. London: Cambridge Univ. Press
- Krasne, F. B. 1976. Invertebrate systems as a means of gaining insight into the nature of learning and memory. In *Neural Mechanisms of Learning and Memory*, ed. M. R. Rosenzweig, E. L. Bennett, pp. 401–29. Cambridge: MIT Press
- Krasne, F. B. 1978. Extrinsic control of intrinsic neuronal plasticity: An hypothesis from work on simple systems. *Brain Res.* 140: 197–216
- Landfield, P. W., Deadwyler, S. A., eds. 1988. *Long-Term Potentiation: From Biophysics to Behavior*. New York: Liss
- Larson, J., Lynch, G. 1986. Induction of synaptic potentiation in hippocampus by patterned stimulation involves two events. *Science* 232: 985–88
- Larson, J., Wong, D., Lynch, G. 1986. Patterned stimulation at the theta frequency is optimal for the induction of hippocampal long-term potentiation. *Brain Res.* 368: 347–50
- Lee, K. S. 1983. Cooperativity among afferents for the induction of long-term potentiation in the CA1 region of the hippocampus. *J. Neurosci.* 3: 1369–72
- Levy, W. B. 1985. Associative changes at the synapse: LTP in the hippocampus. See Levy et al 1985, pp. 5–33
- Levy, W. B., Anderson, J. A., Lehmkuhle, S., eds. 1985. *Synaptic Modification, Neuron Selectivity, and Nervous System Organization*. Hillsdale, NJ: Erlbaum
- Levy, W. B., Burger, B. 1987. *IEEE 1st Int. Conf. on Neural Networks, San Diego*, 4: 11–15. San Diego: IEEE
- Levy, W. B., Desmond, N. L. 1985. The rules of elemental synaptic plasticity. See Levy et al 1985, pp. 105–21
- Levy, W. B., Steward, O. 1979. Synapses as associative memory elements in the hippocampal formation. *Brain Res.* 175: 233–45
- Levy, W. B., Steward, O. 1983. Temporal contiguity requirements for long-term

- associative potentiation/depression in the hippocampus. *Neuroscience* 8: 791-97
- Linsker, R. 1986. From basic network principles to neural architecture: Emergence of spatial-opponent cells. *Proc. Natl. Acad. Sci. USA* 83: 7508-12
- Linsker, R. 1988a. Towards an organizing principle for a layered perceptual network. In *Neural Information Processing Systems*, ed. D. Andersen, pp. 485-94. New York: Am. Inst. Phys.
- Linsker, R. 1988b. Self-organization in a perceptual network. *Computer* 21: 105-17
- Livingston, R. B. 1967a. Brain circuitry relating to complex behavior. See Quarton et al 1967, pp. 499-515
- Livingston, R. B. 1967b. Reinforcement. See Quarton et al 1967, pp. 568-77
- MacDermott, A. B., Mayer, M. L., Westbrook, G. L., Smith, S. J., Barker, J. L. 1986. NMDA-receptor activation increases cytoplasmic calcium concentration in cultured spinal cord neurones. *Nature* 321: 519-22
- Malenka, R. C., Kauer, J. A., Zucker, R. S., Nicoll, R. A. 1988. Postsynaptic calcium is sufficient for potentiation of hippocampal synaptic transmission. *Science* 242: 81-84
- Malinow, R., Miller, J. P. 1986. Postsynaptic hyperpolarization during conditioning reversibly blocks induction of long-term potentiation. *Nature* 320: 529-30
- Mayer, M. L., Westbrook, G. L. 1987. The physiology of excitatory amino acids in the vertebrate central nervous system. *Prog. Neurobiol.* 28: 197-276
- Mayer, M. L., MacDermott, A. B., Westbrook, G. L., Smith, S. J., Barker, J. L. 1987. Agonist- and voltage-gated calcium entry in cultured mouse spinal cord neurons under voltage clamp measured using arsenazo III. *J. Neurosci.* 7: 3230-44
- Mayer, M. L., Westbrook, G. L., Guthrie, P. B. 1984. Voltage-dependent block by Mg^{2+} of NMDA responses in spinal cord neurones. *Nature* 309: 261-63
- McGaugh, J. L. 1988. Modulation of memory storage processes. In *Memory: An Interdisciplinary Approach*, ed. P. R. Solomon, G. R. Goethals, C. M. Kelley, B. R. Stephens. New York: Springer-Verlag. In press
- McGaugh, J. L. 1989. Involvement of hormonal and neuromodulatory systems in the regulation of memory storage. *Annu. Rev. Neurosci.* 12: 255-87
- McNaughton, B. L., Douglas, R. M., Goddard, G. V. 1978. Synaptic enhancement in fascia dentata: Cooperativity among coactive afferents. *Brain Res.* 157: 277-93
- Merzenich, M. M., Nelson, R. J., Stryker, M. P., Cynader, M. S., Schopmann, A., Zook, J. M. 1984. Somatosensory cortical map changes following digit amputation in adult monkeys. *J. Comp. Neurol.* 224: 591-605
- Metherate, R., Tremblay, N., Dykes, R. W. 1987. Acetylcholine permits long-term enhancement of neuronal responsiveness in cat primary somatosensory cortex. *Neuroscience* 22: 75-82
- Miller, M. W., Lee, S. C., Krasne, F. B. 1987. Cooperativity-dependent long-lasting potentiation in the crayfish lateral giant escape reaction circuit. *J. Neurosci.* 7: 1081-92
- Miller, S. G., Kennedy, M. B. 1986. Regulation of brain type II Ca^{2+} /calmodulin-dependent protein kinase by autophosphorylation: A Ca^{2+} -triggered molecular switch. *Cell* 44: 861-70
- Monaghan, D. T., Cotman, C. W. 1985. Distribution of NMDA-sensitive L-[3H]-glutamate binding sites in rat brain as determined by quantitative autoradiography. *J. Neurosci.* 5: 2909-19
- Montorolo, P. G., Kandel, E. R., Schacher, S. 1988. Long-term heterosynaptic inhibition in *Aplysia*. *Nature* 333: 171-74
- Muller, D., Joly, M., Lynch, G. 1988. Contributions of quisqualate and NMDA receptors to the induction and expression of LTP. *Science* 242: 1694-97
- Muller, D., Lynch, G. 1988. Long-term potentiation differentially affects two components of synaptic responses in hippocampus. *Proc. Natl. Acad. Sci. USA* 85: 9346-50
- Nowak, L., Bregestovski, P., Ascher, P., Herbet, A., Prochiantz, A. 1984. Magnesium gates glutamate-activated channels in mouse central neurones. *Nature* 307: 462-65
- Pacelli, G. J., Naghdi, F., Kelso, S. R. 1987. Possible role of inhibition in LTP induction. *Soc. Neurosci. Abstr.* 13: 977
- Palm, G. 1982a. *Neural Assemblies: An Alternative Approach*. Berlin/Heidelberg/New York: Springer-Verlag
- Palm, G. 1982b. Rules for synaptic changes and their relevance for the storage of information in the brain. In *Cybernetics and Systems Research*, ed. R. Trappl, pp. 277-80. Amsterdam: North-Holland
- Palm, G. 1984. Associative networks and cell assemblies. In *Brain Theory*, ed. G. Palm, A. Aertsen, pp. 211-28. Berlin/Heidelberg: Springer-Verlag
- Pearson, J. C., Finkel, L. H., Edelman, G. M. 1987. Plasticity in the organization of adult cerebral cortical maps: A computer simulation based on neuronal group selection. *J. Neurosci.* 7: 4209-23
- Quarton, G. C., Melnechuk, T., Schmitt, F. O., eds. 1967. *The Neurosciences*. New York: Rockefeller Univ. Press

- Racine, R. J., deJonge, M. 1988. Short-term and long-term potentiation in projection pathways and local circuits. See Landfield & Deadwyler 1988, pp. 167–200
- Racine, R. J., Kairiss, E. W. 1987. Long-term potentiation phenomena: The search for the mechanisms underlying memory storage processes. In *Neuroplasticity, Learning, and Memory*, ed. N. W. Milgram, C. M. MacLeod, T. L. Petit, pp. 173–97. New York: Liss
- Racine, R. J., Milgram, N. W., Hafner, S. 1983. Long-term potentiation phenomena in the rat limbic forebrain. *Brain Res.* 260: 217–31
- Ranck, J. B. 1964. Synaptic “learning” due to electroosmosis: A theory. *Science* 144: 187–89
- Rauschecker, J. P., Hahn, S. 1987. Ketamine-xylazine anaesthesia blocks consolidation of ocular dominance changes in kitten visual cortex. *Nature* 326: 183–85
- Reiter, H. O., Stryker, M. P. 1988. Neural plasticity without postsynaptic action potentials: Less-active inputs become dominant when kitten visual cortical cells are pharmacologically inhibited. *Proc. Natl. Acad. Sci. USA* 85: 3623–27
- Rescorla, R. A., Wagner, A. R. 1972. A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In *Classical Conditioning II*, ed. A. H. Black, W. F. Prokasy, pp. 64–99. New York: Appleton-Century-Crofts
- Reymann, K. G., Malisch, R., Schulzeck, K., Brodemann, R., Ott, T., et al. 1985. The duration of long-term potentiation in the CA1 region of the hippocampal slice. *Brain Res. Bull.* 15: 249–55
- Rinaldi, P. C., Ganong, A. H., Brown, T. H. 1987. Low-frequency synaptic depression recorded intracellularly from hippocampal granule cells. *Soc. Neurosci. Abstr.* 13: 977
- Rose, G. M., Dunwiddie, T. V. 1986. Induction of hippocampal long-term potentiation using physiologically patterned stimulation. *Neurosci. Lett.* 69: 244–48
- Rosenblatt, F. 1967. Recent work on theoretical models of biological memory. In *Computer and Information Sciences—II*, ed. J. T. Tou, pp. 33–56. New York: Academic
- Sastry, B. R., Goh, J. W., Auyeung, A. 1986. Associative induction of posttetanic and long-term potentiation in CA1 neurons of rat hippocampus. *Science* 232: 988–90
- Schmajuk, N. A., Moore, J. W. 1988. The hippocampus and the classically conditioned nictitating membrane response: A real-time attentional-associative model. *Psychobiology* 16: 20–35
- Sejnowski, T. J. 1977a. Strong covariance with nonlinearly interacting neurons. *J. Math. Biol.* 4: 303–21
- Sejnowski, T. J. 1977b. Statistical constraints on synaptic plasticity. *J. Theor. Biol.* 69: 385–89
- Sejnowski, T. J. 1981. Skeleton filters in the brain. In *Parallel Models of Associative Memory*, ed. G. E. Hinton, J. A. Anderson, pp. 49–82. Hillsdale, NJ: Erlbaum
- Sejnowski, T. J., Churchland, P. S. 1989. Brain and cognition. In *Foundation of Cognitive Science*, ed. M. I. Posner. Cambridge: MIT Press. In press
- Sejnowski, T. J., Koch, K., Churchland, P. S. 1988. Computational neuroscience. *Science* 241: 1299–1306
- Sejnowski, T. J., Tesauero, G. 1989. The Hebb rule for synaptic plasticity: Algorithms and implementations. See Byrne & Berry 1989, pp. 94–103
- Shatz, C. J., Stryker, M. P. 1978. Ocular dominance in layer IV of the cat’s visual cortex and the effects of monocular deprivation. *J. Physiol.* 281: 267–83
- Sillito, A. M. 1987. Synaptic processes and neurotransmitters operating in the central visual system: A systems approach. See Edelman et al 1987, pp. 328–72
- Singer, W. 1988. Ontogenic self-organization and learning. In *Brain Organization and Memory: Cells, Systems, and Circuits*, ed. J. McGaugh. New York: Oxford Univ. Press. In press
- Singer, W., Kleinschmidt, A., Bear, M. F. 1986. Infusion of an NMDA receptor antagonist disrupts ocular dominance plasticity in kitten striate cortex. *Soc. Neurosci. Abstr.* 12: 786
- Stanton, P. K., Sejnowski, T. J. 1989. Associative long-term depression in the hippocampus: Induction of synaptic plasticity by Hebbian covariance. *Nature* 339: 215–18
- Staubli, U., Lynch, G. 1987. Stable hippocampal long-term potentiation elicited by “theta” pattern stimulation. *Brain Res.* 435: 227–34
- Stent, G. S. 1973. A physiological mechanism for Hebb’s postulate of learning. *Proc. Natl. Acad. Sci. USA* 70: 997–1001
- Sutton, R. S., Barto, A. G. 1981. An adaptive network that constructs and uses an internal model of its world. *Cogn. Brain Theor.* 4: 217–46
- Tanzi, E. 1893. I fatti e la induzioni nell’odierna istologia del sistema nervoso. *Riv. Sper. Freniatr. Med. Leg. Alienazioni Ment. Soc. Ital. Psichiatria* 19: 419–72
- Tesauro, G. 1986. Simple neural models of classical conditioning. *Biol. Cybern.* 55: 187–200
- Teyler, T. J., DiScenna, P. 1987. Long-term

- potentiation. *Annu. Rev. Neurosci.* 10: 131-61
- Tsien, R. Y., Zucker, R. S. 1986. Control of cytoplasmic calcium with photolabile tetracarboxylate 2-nitrobenzhydrol chelators. *Biophys. J.* 50: 843-53
- Volper, D. J., Hampson, S. E. 1987. Learning and using specific instances. *Biol. Cybern.* 57: 57-71
- von den Malsburg, C. 1973. Self-organization of orientation sensitive cells in the striate cortex. *Kybernetik* 14: 85-100
- von den Malsburg, C., Bienenstock, E. 1986. Statistical coding and short-term synaptic plasticity: A scheme for knowledge representation in the brain. In *Disordered Systems and Biological Organization*, ed. E. Bienenstock, F. Fogelman Soulié, G. Weisbuch, pp. 247-72. Berlin/Heidelberg: Springer-Verlag
- Walters, E. T., Byrne, J. H. 1985. Long-term enhancement produced by activity-dependent modulation of *Aplysia* sensory neurons. *J. Neurophysiol.* 5: 662-72
- Wickens, J. 1988. Electrically coupled but chemically isolated synapses: Dendritic spines and calcium in a rule for synaptic modification. *Prog. Neurobiol.* 31: 507-28
- Wiesel, T. N., Hubel, D. H. 1963. Single-cell responses in striate cortex of kittens deprived of vision in one eye. *J. Neurophysiol.* 26: 1003-17
- Wiesel, T. N., Hubel, D. H. 1965. Comparison of the effects of unilateral and bilateral eye closure on cortical unit responses in kittens. *J. Neurophysiol.* 28: 1029-40
- Wigström, H., Gustafsson, B. 1983. Large long-lasting potentiation in the dentate gyrus *in vitro* during blockade of inhibition. *Brain Res.* 275: 153-58
- Wigström, H., Gustafsson, B., Huang, Y.-Y., Abraham, W. C. 1986. Hippocampal long-term potentiation is induced by pairing single afferent volleys with intracellularly injected depolarizing pulses. *Acta Physiol. Scand.* 126: 317-19
- Williams, S., Johnston, D. 1988. Muscarinic depression of long-term potentiation in CA3 hippocampal neurons. *Science* 242: 84-87
- Wood-Jones, F., Porteus, S. D. 1928. *The Matrix of the Mind*. Honolulu: Mercantile
- Woodworth, R. S. 1921. *Psychology: A Study of Mental Life*. New York: Holt
- Zador, A., Koch, C., Brown, T. H. 1990. Biophysical model of a Hebbian synapse. *Proc. Int. Joint Conf. Neural Networks 1990*. In press
- Ziegler, H. E. 1900. Theoretisches zur Tierpsychologie und vergleichender Neurophysiologie. *Biol. Central.* 20: In press
- Zucker, R. S. 1972. Crayfish escape behavior and central synapses. II. Physiological mechanisms underlying behavioral habituation. *J. Neurophysiol.* 35: 621-37
- Zucker, R. S., Bruner, J. 1977. Long-lasting depression and the depletion hypothesis at crayfish neuromuscular junctions. *J. Comp. Physiol.* 121: 223-40